The Master Equation from four axioms: the field equation of the minimal twelve-dimensional geometry

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Abstract

Einstein's field equations relate the curvature of a space to the matter in it and to the energy of empty space, and in a century of use they have described gravity and left the other forces, the existence and structure of matter, and the direction of time to be supplied from outside. The first paper of this series derived, from four sentences a child can verify, a unique twelve-dimensional geometry. We show here that the same four sentences, with the same principle of minimality, derive a field equation on that geometry, and that the equation is Einstein's with one additional term. Two of the axioms, closure and order, give Einstein's equation with its cosmological and matter terms. The third, that the universe grows and decays as a function of itself, gives the additional term: the exponential of growth and decay has two sheets, the sheets are one field ψ, a section of the Clifford module of the geometry, with shadow $\bar{\psi}$, their even combination $\omega = \psi + \bar{\psi}$ and the unique lowest-order even scalar vanishing both where the sheets coincide and where growth opposes decay, $\Phi_{\mathrm{QC}} = \omega|\psi - \bar{\psi}|^2$, enter the action through one coupling ζ, and the variation gives $G_{MN} = 8\pi G_{\mathrm{eff}} T_{MN} + \zeta T_{MN}^{\Phi_{\mathrm{QC}}} + \Lambda_{\mathrm{eff}} g_{MN}$. The fourth axiom, the drive, makes the two sheets mirror images relative to the flow of time rather than one sheet written backwards, so that the term is physical and the arrow of time and the distinctness of matter from its shadow are the same fact; no Lorentzian structure is assumed. The coupling is derived from the restriction of the equation to the base of the geometry as the exponent by which the warp falls across a half-period, $\zeta = \chi(K_8)\pi = 12\pi$. The derivation is minimal rather than unique: at four named points the smallest option the axioms admit is taken, and each is tagged. The field ω vanishes at the locked phase by the second axiom and at the uncorrelated phase by the fourth, and the coupling is largest at the sheet angle $\arctan\sqrt{2}$.

1 What is observed, and what this paper derives

The purpose of this paper is to derive the field equation of the Master Equation programme from the four axioms of M1, on the geometry M1 showed those axioms select, and to show that the equation is Einstein's with one additional term, produced by the one axiom Einstein's theory does not contain.

1.1 The result in plain words

M1 began from four observations, taken as axioms in the simplest words that carry them, and we restate them as M1 did.

Axiom 1 (closure)**.** The universe closes on itself, so that it can be aware of itself.

Axiom 2 (growth and decay)**.** The universe grows and decays as a function of itself.

Axiom 3 (order)**.** The order of events matters, so that each instance is unique and cannot be foretold.

Axiom 4 (drive)**.** The universe is in constant change; it never stands still.

Read as conditions on a shape, with every point at which a word acquires a mathematical meaning isolated as a numbered reading, and with the rule that no unforced structure is added, these four picked out one shape, twelve-dimensional, described in M1. This paper asks what rule of motion the same four sentences impose on that shape.

The answer can be stated before any mathematics. Einstein's equation says that the curvature of space is produced by two things: the matter in it, and the energy that empty space has of its own. Our equation says the same, with one more source of curvature added: the difference between growth and decay, weighted by how coherently the two are held together. Written as a formula it is Einstein's equation with one extra term on the right-hand side, and the extra term has one number in front of it, which we compute to be 12π.

Which observation produces which part of the equation is the content of the paper, and it is worth saying at once. That the universe closes on itself is what makes an equation of motion exist at all, and it supplies the energy of empty space; that the order of events matters is what supplies matter, in the form of the spinor fields that carry non-commutativity, together with the forces that act between them. These two observations alone give Einstein's equation, with its matter and its cosmological constant. That the universe grows and decays is the observation Einstein's theory does not contain, and it is the one that produces the additional term: the growth law has two faces, growth and decay, and the field written on those two faces is a new source of curvature. That the universe never stands still is what makes the two faces genuinely two rather than one face read in a mirror; it is the source of the arrow of time, and it is why matter is distinct from its shadow. In the programme that follows this paper, the second observation is also where the masses, the generations and the mixings of matter come from, so that what was missing from a theory of four forces, an account of why there is matter and why it does not run backwards, is supplied by the two observations the earlier theory left out. We do not derive those consequences here; we derive the equation and its coupling.

1.2 What the mathematics is doing

For a reader meeting this for the first time, the argument has three moves, and each can be said in ordinary words.

The first move is to write down a single number, the *action*, for every way the shape and the fields on it could possibly be arranged, and then to pick out the arrangement for which that number is *stationary*. Stationary means this: if you alter the arrangement by a tiny amount in any direction — bend the shape slightly here, raise the field slightly there — the number does not change to first order. It is the same idea as finding the bottom of a valley by looking for the place where the ground is level: at the lowest point, a small step in any direction neither raises nor lowers you. For a function of one variable this is the condition that its derivative be zero; the action is a function of infinitely many variables (the value of every field at every point), and the stationary condition is that its derivative with respect to each of them be zero at once. Working out what those conditions say is a mechanical calculation, and the result is the field equation. This is how every field equation in physics is obtained, Einstein's included: the equation is not guessed, it is the condition that the action be stationary. The first axiom is what makes the move possible, because a shape that closes on itself has no edge over which the small alterations could leak away and spoil the count.

The content of the paper is then entirely in *which terms the action is allowed to contain*. We allow nothing the axioms do not force, and where the axioms leave a choice we take the smallest option, as M1 did; the paper names each such choice. There are three kinds of term: the curvature of the shape itself, the matter fields and the force field that the third axiom supplies, and the field written on the two faces of the growth law that the second axiom supplies.

The second move is to identify that last field. The growth law $w = w_0 e^{\kappa s}$ has two solutions, growth and decay, exchanged by an operation the second axiom asks for. We show that this operation is the reversal of the drive of the fourth axiom together with a quarter-turn of the

internal fibre, that it reverses the orientation of the shape, and hence that the two solutions, written as one field ψ and its shadow $\bar{\psi}$, have opposite handedness. The even combination $\omega = \psi + \bar{\psi}$ measures how far the two faces agree; the odd combination $\psi - \bar{\psi}$ measures how far they are held apart. The one scalar built from them that respects the exchange and vanishes both when the faces coincide and when growth exactly opposes decay is $\Phi_{\mathrm{QC}} = \omega|\psi - \bar{\psi}|^2$, and the shape couples to the two faces through it and through nothing else.

The third move is the computation. Varying the action gives Einstein's equation with the one extra term; the extra term is the stress tensor of a scalar field, a standard object, multiplied by the coupling ζ. To find ζ we restrict the equation to the two-dimensional base of the internal shape, where the growth law appears as a warp that falls off across the base, and we show that across one half of the base the warp falls by exactly $N\pi$, where N is the number of units of flux the shape carries and π is the angle missing at each of the four corners of the pillowcase. The number of flux units is the Euler characteristic of the internal shape, twelve, so $\zeta = 12\pi$.

1.3 What this paper proves, and how it is organised

The results, in the order they are proved, are these. The sheets of A2 are one field ψ, a section of the Clifford module $\mathbb{H}^{32}$ that A3 forces, with shadow $\bar{\psi}$ obtained by the flow-reversing involution; $\omega = \psi + \bar{\psi}$ and $\psi - \bar{\psi}$ are the Coherent Field and the sheet variable, and $\Phi_{\mathrm{QC}} = \omega|\psi - \bar{\psi}|^2$ is the unique lowest-order even scalar vanishing both where the sheets coincide and where growth opposes decay (Section 2). The involution is the quarter-turn flow reversal τ with τ^2 the fold, it reverses orientation, and $\bar{\psi}$ has the opposite handedness to ψ (Section 2.5). The action on M_{12} that the axioms and the standing minimality principle admit has one gravitational term, the induced connection of the Spin^c line, and the dilaton coupled through Φ_{QC} alone; its variation is the Master Equation, Einstein's equation with the one additional term, and the dilaton's equation is $\zeta\Box\Phi_{\mathrm{QC}} = V'$ (Section 3). The restriction of the equation to the base is the two-dimensional system in which the warp falls by $N\pi$ across a half-period, $N = \chi(K_8) = 12$, so $\zeta = 12\pi$ (Section 4). The Coherent Field vanishes at the locked phase by A2 and at the uncorrelated phase by A4, the coupling is largest at $\arctan\sqrt{2}$, and ω keeps one sign on the solutions (Section 5).

The paper is organised as follows. Section 2 defines the variables, states the fifth reading of the series, that the metric is positive-definite and time is the drive's orbit parameter, and states the standing principle of minimality. Section 2.5 proves the handedness theorem. Section 3 writes the action and derives the equation, saying for each term which axiom produces it. Section 4 derives the value of ζ. Section 5 derives the shape of ω. Section 6 says what the equation dictates for the physics and what remains open. As in M1, this introduction and the opening of each section are intended to be followed by a first-year student; from Section 2.5 the register is that of a formal paper, every load-bearing step is proved or reduced to a cited classical result or to a stated reading, the smallest case is worked by hand, and the finite and arithmetic parts are certified in Lean (Appendix A). Rigour is tagged [R], [D], [M], [G] as in M1. The deflationary reading of the paper is stated where it belongs: the equation is derived from the action, and the action is fixed by the axioms together with four minimal choices, each named (Sections 3.5 and 4.9); a reader who rejects minimality as a principle has a family of actions, not an equation. Every symbol of M1 keeps its meaning; the symbols introduced here are tabulated in Appendix B.

2 The variables

The geometry is fixed by M1: $M_{12} = (S^1 \times \Sigma) \times K_8$ with $\Sigma = S^3$ and $K_8 = [(\mathbb{CP}^2 \times S^2) \times_w (T^2/\mathbb{Z}_2)]^{\mathrm{Spin}^c}$, the fibre T^2 carrying the involution of A2 (the fold on the pillowcase; refined in Section 2.5) and the Fibonacci drive M_0 of A4, with corners $P_1, \ldots, P_4$ labelled as in M1

Appendix B. We keep every convention of M1 unchanged; in particular the fibre coordinates are $(\theta_\psi, \theta_\omega)$ with half-periods (L_ψ, L_ω). In this paper ψ and ω also name *fields*. The two uses are distinguished by position: a subscript $\theta_\psi, \theta_\omega, L_\psi, L_\omega, k_\omega$ is a coordinate label inherited from M1; a bare ψ, $\bar{\psi}$ or ω is a field defined in this section. The cumulative table of symbols is Appendix B.

2.1 The fifth reading

M1 numbered four readings by which the four sentences became conditions, and its abstract already describes the geometry as compact and Riemannian without making that a numbered reading. This paper makes it one, and attaches it to every result that depends on it.

Definition 2.1 (the metric signs no direction)**.** The metric of M_{12} is positive-definite. Time is not a direction the metric distinguishes by a sign; it is the orbit parameter s of the flow of A4 on $T \times F$. A derivative "along time" is a derivative along that flow, and every statement below that reads as a statement about time is a statement along the flow.

The reading is the geometric form of a sentence as plain as M1's four: the universe does not label its directions. It is also the reading under which A1 is unproblematic, since a compact closed manifold with a signed direction would contain closed time-like curves, and under which the spectra of the operators of the programme are real and discrete. What singles out a direction is the drive, and Section 2.5 shows that the drive singles it out with an orientation. [D]

The standing principle. M1 chose among geometries by a minimality principle (M1 Definition 3.2), and every paper of the series applies the same principle to every datum the axioms leave open: terms in an action, connections, couplings, readings. Where a choice exists we take the minimal option and tag the result [D] with minimality as its hypothesis, on the same footing as M1's own minimality results; and where the minimal option fails — a contradiction, an empty solution set, a conflict with a result already established — we say so where it happens rather than enlarge the choice silently. The present paper applies the principle three times (Section 3.1), and names each instance.

2.2 One complex field from two sheets

Axiom A2 supplies a growth law $w = w_0 e^{\kappa s}$ along the warped direction and an involution ι with $\iota^* \kappa = -\kappa$ (M1 Definition 2.2). The two signs of κ are the two *sheets* of the exponential, growth and decay, exchanged by ι. M1 identifies ι on the fibre algebra: at the drive-fixed corner $P_1 = C_0$ it is conjugation on $\mathbb{H}$, whose commutative fixed subalgebra containing the selected unit is $\mathbb{C}_i$ (M1 Propositions 5.4 and 5.6); at the three cycled corners $P_3 = C_i$, $P_2 = C_j$, $P_4 = C_k$ it is the inner automorphism by the surviving unit, and the drive's three-cycle $\mathrm{Ad}_w : i \to j \to k$ carries one corner's reading to the next (M1 Proposition 5.6, certificate `L2_Ad_w_cycles`).

Before the definition, three facts it rests on, each from M1, stated in words. First, the third axiom says the fibre algebra — the quaternions, by M1 Proposition 5.2 — acts on spinors (M1 Definition 2.5); so whatever field the two sheets of the growth law make, it is a spinor-type object, and its values live in the space on which the quaternions act. Second, that space is fixed by the dimension: in twelve dimensions the Clifford algebra is the algebra of 32×32 quaternionic matrices, by the eightfold periodicity of Bott (twelve is four more than eight, and the four-dimensional case is $\mathbb{H}(2)$, so the twelve-dimensional case is $\mathbb{H}(2)$ enlarged sixteenfold); the space it acts on irreducibly is therefore thirty-two quaternions, which is sixty-four complex numbers or one hundred and twenty-eight real ones (`L7_module_dims`). Third, the operation that exchanges the two sheets is the flow reversal τ of Section 2.5: reverse the drive and turn the fibre by a quarter. Done twice it is the fold of the torus that makes the pillowcase, so on the pillowcase itself it is an involution.

Definition 2.2 (the field ψ and its shadow)**.** Let $M \cong \mathbb{H}^{32}$ be the irreducible module of the Clifford algebra of M_{12}. A *sheet field* is a smooth section ψ of M: at every point of the shape, thirty-two quaternions $\psi_1, \ldots, \psi_{32}$, varying smoothly from point to point. Let $\hat{\tau}$ be the lift of the flow reversal τ to M — the operation that carries a spinor at a point to a spinor at the reversed point, and which, because τ reverses orientation, is an odd element of the Pin group. The *shadow* of ψ is the sheet field

$$\bar{\psi} \;\equiv\; \hat{\tau}\psi\,, \tag{1}$$

the field as seen after the exchange of growth and decay. Applying $\hat{\tau}$ twice returns ψ carried round the fold, which on the pillowcase is ψ itself up to the sign ± 1 that M1 Corollary 7.4 assigns to each corner. So on the pillowcase every sheet field splits into an even part and an odd part under the exchange,

$$\psi \;=\; \tfrac{1}{2}(\psi + \bar{\psi}) \;+\; \tfrac{1}{2}(\psi - \bar{\psi})\,,$$

the first unchanged by exchanging growth and decay, the second reversed by it. Finally, at a corner C_u of the pillowcase the quaternions collapse to the complex numbers $\mathbb{C}_u$ fixed by the corner reading (M1 Propositions 5.4 and 5.6); the *corner reading* of a sheet field is its image there, thirty-two complex numbers instead of thirty-two quaternions, and it is in that reading that the phase of Section 5 is defined.

The two sheets of A2 are therefore one field: its $\hat{\tau}$-even part is what growth and decay share, its $\hat{\tau}$-odd part is what distinguishes them. That the field is module-valued rather than real or complex is not an extra choice: M1 Proposition 7.3 reads the spinor data of A3 as the Spin^c module, and the selected imaginary unit of Proposition 7.2 is the direction in $\mathrm{Im}\,\mathbb{H}$ along which the even part carries its imaginary component. In the corner reading each component is a complex number $x + iy$ in $\mathbb{C}_u$ and $\hat{\tau}$ sends $y \mapsto -y$. [D]

2.3 The even and odd variables, and the dilaton

Definition 2.3 (ω and the odd variable)**.**

$$\omega \;\equiv\; \psi + \bar{\psi}\,, \qquad \psi - \bar{\psi}\,, \tag{2}$$

both sections of M. The first is the even part of the sheet field doubled, the second the odd part doubled. ω is the Coherent Field of the programme: it measures how much the two sheets agree. It is quaternion-valued, and its imaginary part lies along the selected unit; since the odd variable will enter the dilaton only through its modulus, a non-negative real number, ω is the only one of the two that can give the dilaton a direction (Section 2.3). In the corner reading, where each component is a complex number, $\omega = 2\,\mathrm{Re}\,\psi$ and $\psi - \bar{\psi} = 2i\,\mathrm{Im}\,\psi$.

Two properties of ω are fixed at the level of the definition and are the inputs to the next proposition. First, ω is the even *combination* of the sheets and not their modulus: when growth stands exactly against decay, $\psi = -\bar{\psi}$, nothing in the geometry changes and $\omega = 0$. The modulus $|\psi|$ does not vanish there; the combination does. Second, when the two sheets coincide, $\psi = \bar{\psi}$, there is no held distinction between them, and we require that whatever couples the sheets to the curvature vanish: this is the motivated condition that the coupling is a coupling *between* the sheets, not a coupling of either sheet alone. [M]

Proposition 2.4 (the lowest even invariant)**.** *Write $E = \frac{1}{2}(\psi + \bar{\psi})$ for the even part and $O = \frac{1}{2}(\psi - \bar{\psi})$ for the odd part of a sheet field. Consider scalars built by multiplying powers of E and powers of $|O|$ together, and ask for those that are unchanged when growth and decay are exchanged, and that vanish both when the sheets coincide ($O = 0$) and when growth exactly opposes decay ($E = 0$). Every such scalar has total degree at least three, and the one of degree three is, up to a real constant, $E\,|O|^2$. In the programme's normalisation it is the dilaton*

$$\Phi_{\mathrm{QC}} \;=\; \omega\,|\psi - \bar{\psi}|^2\,, \tag{3}$$

taken component by component in M: the α-th component of Φ_{QC} is ω_α times the squared modulus of $(\psi - \bar{\psi})_\alpha$.

Proof. We work in the corner reading, where each component is a complex number $x + iy$ with $x = \operatorname{Re}\psi$ the even part and $y = \operatorname{Im}\psi$ the odd part; the general case is the same argument in each component. A monomial $x^a y^b$ changes sign under the exchange exactly when b is odd, because the exchange sends y to $-y$ and leaves x alone; so the even monomials are those with b even. The monomial vanishes identically on the locus $y = 0$ only if y actually appears, that is $b \geq 1$, and on the locus $x = 0$ only if x appears, $a \geq 1$. Put the three conditions together: b is even and at least one, so $b \geq 2$; $a \geq 1$; hence $a + b \geq 3$, with equality only when $a = 1$ and $b = 2$. The unique degree-three candidate is therefore xy^2. In the original variables $x = \omega/2$ and $y^2 = |\psi - \bar{\psi}|^2/4$, so $xy^2 = \frac{1}{8}\,\omega|\psi - \bar{\psi}|^2$, which is (3) up to the constant $\frac{1}{8}$. The classification of exponents, the two vanishing criteria and the parity table of ω, $\psi - \bar{\psi}$, $|\psi - \bar{\psi}|^2$ and Φ_{QC} are certified in `M2_L1_parity.lean` (`L1_unique`, `L1_vanish_Im`, `L1_vanish_Re`, `L1_table_*`). [R] □

One remark on the module. The factor $|\psi - \bar{\psi}|^2$ is a non-negative real number in each component, so it carries no direction in the quaternions: whatever direction Φ_{QC} has, it has because ω has it. In particular Φ_{QC} has an imaginary part only through ω, and ω is the unique carrier of the imaginary direction (`L7_imaginary_square_real`, which certifies the supporting fact that the square of an imaginary quaternion is the negative of its squared modulus). The real factor also commutes with everything, so no question of the order of factors arises. The quaternionic structure that the fibre paper of the series reads as the $Sp(1)$ flux on S^2 is carried by the direction of ω; this paper does not use it. [R]

We record what the proposition does and does not decide. Parity alone does not choose $\psi + \bar{\psi}$ over the modulus $|\psi|$ as the even factor: both are unchanged by the exchange. The choice is the first property above, that ω vanishes when growth stands against decay; with the modulus in its place Φ_{QC} would not vanish there. The proposition then shows that once the even variable is ω and the two vanishing conditions are imposed, there is no further freedom at lowest order.

The smallest case by hand: the drive-fixed corner P_1 is fixed by τ (Section 2.5), so $\bar{\psi}(P_1) = \psi(P_1)$ up to the corner sign, so $\operatorname{Im}\psi(0) = 0$ and $\Phi_{\mathrm{QC}}(P_1) = 0$: the dilaton vanishes at the real corner, where there is one sheet and nothing to hold apart.

2.4 What is varied, and over what

The field equation of this paper is obtained by variation and reduction. We state the scheme once so that each later section can cite it.

Definition 2.5 (the variational data)**.** (i) *The domain* is M_{12} of M1: compact, connected, without boundary (A1, M1 Definition 2.1).

(ii) *The fields varied* are the twelve-dimensional metric g, with the A2 warp $w = e^A$ depending on the base coordinates $(\theta_\psi, \theta_\omega)$ (M1 Definition 2.2(i)); the dilaton Φ_{QC} of Section 2.3, a section of the Clifford module M on M_{12} with coupling ζ (Section 3 states how its kinetic term arises from the sheets); the Spin^c spinor of A3 on the bundle with determinant line $L = \mathcal{O}(3)$ (M1 Proposition 7.3); and the Spin^c connection.

(iii) *The reduction* integrates over K_8 in two stages: over the six-dimensional fibre $\mathbb{CP}^2 \times S^2$, leaving a two-dimensional system on the pillowcase base $T^2/\mathbb{Z}_2$ (Section 4); and over the base, leaving the four-dimensional equation on $S^1 \times \Sigma$ (Section 3). The field equation of this paper is the twelve-dimensional one, on M_{12}; the reduction is how its consequences on the base and on the external block are read.

(iv) *The scale* of M_{12} is the declared input of M1 (Theorem 8.1, Proposition 8.2) and is not varied.

2.5 The arrow, and why the shadow is not the image

Nothing so far distinguishes ψ from $\bar{\psi}$ except a sign under the fold. If that were all, the second sector would be the first written backwards, and $|\psi - \bar{\psi}|^2$ would measure a convention. This subsection shows that it is not all: the operation relating the two sectors reverses the orientation of the geometry, so that $\bar{\psi}$ is the mirror of ψ relative to the flow of A4, and the two carry opposite handedness. The argument uses no sign in the metric. Under the fifth reading of Section 2.1 the metric of M_{12} is positive-definite and signs no direction; the distinction between the sectors is made by the drive.

The Sol block. A4 places a flow on $T \times F$ whose return map is the drive $M_0 = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ (M1 Theorem 4.2). The flow is therefore the suspension of M_0: the three-manifold obtained from $T^2 \times [0,1]$ by identifying $(x, 1)$ with $(M_0 x, 0)$, with the flow ∂_s along the interval. A torus bundle over a circle with hyperbolic monodromy carries the Sol geometry [10], and in the eigenbasis of M_0 the Sol metric is

$$\mathrm{d}\ell^2 \;=\; e^{2\kappa s}\,\mathrm{d}x_u^2 + e^{-2\kappa s}\,\mathrm{d}x_s^2 + \mathrm{d}s^2\,, \qquad \kappa = \log\lambda = \log\varphi^2\,, \tag{4}$$

where x_u, x_s are coordinates along the unstable and stable eigenvectors $v_u = (\varphi, 1)$ and $v_s = (-1, \varphi)$ of M_0, with eigenvalues φ^2 and $\varphi^{-2} = 2 - \varphi$ (`L4_eigen`). This is the factor S^1_{Sol} in the name of the geometry, and it is A2's law seen in A4's drive: one direction grows as $e^{\kappa s}$ and the other decays as $e^{-\kappa s}$, at the rate fixed by the trace-three theorem. The two sheets of the exponential are the two eigen-directions of the drive. [D]

Reversing the flow. A symmetry of the Sol block that reverses the flow has the form $(x, s) \mapsto (Ax, -s)$ with $A \in GL(2, \mathbb{Z})$, and it is well defined on the suspension iff $AM_0A^{-1} = M_0^{-1}$: reversing time inverts the monodromy, and A must carry the inverted monodromy back. Such an A necessarily exchanges the two eigenlines, since it sends the eigenvalue λ to λ^{-1}. Two kinds exist.

Proposition 2.6 (flow reversals)**.** *Let $A \in GL(2, \mathbb{Z})$ satisfy $AM_0A^{-1} = M_0^{-1}$.*

(i) If $A^2 = I$ then $\det A = -1$, and the reversal $(Ax, -s)$ preserves the orientation of the Sol block.

(ii) If $\det A = +1$ then $A^2 = -I$; the reversal has order four, its square is the fold $x \mapsto -x$, and it reverses the orientation of the Sol block. The quarter-turn $S = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ is such an A: $SM_0S^{-1} = M_0^{-1}$, $S^2 = -I$, $Sv_u = -v_s$, $Sv_s = v_u$.

Proof. (i) Write the matrix as $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ with integer entries. Squaring it and setting the result equal to the identity gives four equations; the two off-diagonal ones read $b(a+d) = 0$ and $c(a+d) = 0$. There are two cases. If $a+d \neq 0$ then b and c are both zero and A is diagonal; but a diagonal matrix conjugating M_0 to its inverse would have to send the top-left entry 2 of M_0 to the top-left entry 1 of M_0^{-1}, which forces $a = 0$, contradicting $a^2 = 1$ from the diagonal of $A^2 = I$. So $a+d = 0$, that is $d = -a$, and then the determinant is $ad - bc = -a^2 - bc = -(a^2 + bc) = -1$, using the top-left entry $a^2 + bc = 1$ of $A^2 = I$. The reversal $(x, s) \mapsto (Ax, -s)$ then has determinant $\det A \cdot (-1) = +1$: it preserves orientation (`L4_involution_det`). (ii) Any A that exchanges the two eigenlines has zero diagonal in the eigenbasis, hence zero trace, and by Cayley–Hamilton $A^2 = (\operatorname{tr} A)A - (\det A)I = -I$ when $\det A = 1$. The quarter-turn S is such an A: multiplying out, $SM_0S^{-1} = M_0^{-1}$ and $S^2 = -I$ (`L4_S_conj`, `L4_S_sq_fold`); it sends $(\varphi, 1)$ to $(1, -\varphi)$, which is minus the stable eigenvector, and $(-1, \varphi)$ to $(\varphi, 1)$ (`L4_swap_lines`); and the reversal built from it has determinant $\det S \cdot (-1) = -1$, so it reverses orientation (`L4_tau_reverses`). [R] □

The smallest case by hand: S sends $(\varphi, 1)$ to $(1, -\varphi) = -(-1, \varphi)$, the unstable line to the stable line. Running the flow backwards and turning the fibre by a quarter exchanges growth and decay exactly.

The involution of A2 is the quarter-turn reversal. A2(iii) asks for an involution of the warped piece with $\iota^*\kappa = -\kappa$. On the Sol block the only operations that send κ to $-\kappa$ are the flow reversals of Proposition 2.6, and among them only case (ii) reverses the orientation of the geometry. We take

$$\tau : (x, s) \longmapsto (Sx, -s)\,, \qquad \tau^2 = \text{the fold}, \tag{5}$$

as the involution of A2. On the covering torus τ has order four; on the pillowcase $T^2/\{\pm 1\}$, which is where M1 reads A2, τ^2 is the identity and τ is the involution M1 requires. M1's identification of ι with the fold is therefore the statement about τ^2; the present paper refines it to τ, whose square it is. [D]

Handedness. A reflection of one direction of a Riemannian manifold lifts to the spinor bundle as an element of the Pin group lying in the odd component, and every odd element anticommutes with the chirality operator $\Gamma = \Gamma_1 \cdots \Gamma_{12}$ [6]. The lift $\hat{\tau}$ of τ reverses orientation (Proposition 2.6), so $\hat{\tau}\Gamma = -\Gamma\hat{\tau}$.

Theorem 2.7 (the two sectors have opposite handedness)**.** *Let ψ be a sheet field of definite chirality, $\Gamma\psi = \epsilon\psi$, and $\bar{\psi} \equiv \hat{\tau}\psi$ with $\hat{\tau}$ the Pin lift of τ acting on the module M. Then $\Gamma\bar{\psi} = -\epsilon\bar{\psi}$. Equivalently: the frame (v_u, v_s, ∂_s) in which ψ grows along the flow is carried by τ to a frame of the opposite orientation, $(-v_s, v_u, -\partial_s)$, in which $\bar{\psi}$ grows along the flow.*

Proof. $\Gamma\bar{\psi} = \Gamma\hat{\tau}\psi = -\hat{\tau}\Gamma\psi = -\epsilon\hat{\tau}\psi = -\epsilon\bar{\psi}$. The frame statement is `L4_swap_lines` together with $s \mapsto -s$; the determinant of the change of frame is -1. [R] □

This is the content of the word "shadow". The second sector is not the first reflected in a mirror placed in space; it is the first reflected in the flow. Reversing the drive exchanges growth with decay, and the geometry records which is which: that record is the arrow of time, and the arrow is what keeps ψ and $\bar{\psi}$ apart. Had the flow no stable and unstable direction, running it backwards would be indistinguishable from running it forwards, there would be no arrow, and ψ and $\bar{\psi}$ would be one sector written twice. A4's positive entropy, which M1 used to force the hyperbolic drive, is the same fact as the distinctness of the two sectors.

We record what has been used. The fifth reading (positive-definite metric) makes the chirality operator a Riemannian one and the Pin argument available in the form stated; no Lorentzian structure enters. The identification of A2's involution with τ rather than with its square is the one reading of this subsection, tagged [D] above. The spinor lift of τ is a standard construction once τ is an isometry of the Sol metric, which it is by (4) since S exchanges x_u and x_s while $s \mapsto -s$ exchanges the exponents.

The warped direction. A2's warp and the Sol exponential of (4) share the flow direction s. They do not share a rate: the drive stretches the fibre torus by φ^2 per circuit, a theorem of A4, while the warp of the six-dimensional fibre over the base carries the exponent fixed in Section 4, a theorem of the flux. Two forced numbers are not an unforced datum under M1's first minimality criterion, and the ratio of the two rates is a pure number of the geometry with no input. Whether the base coordinate θ_ω of M1 is the flow coordinate s is settled by the reduction of Section 3 and is not assumed here. [M]

3 The field equation

The purpose of this section is to write the action on M_{12} that the axioms admit, to vary it, and to display the field equation that results. In plain terms: we write down the one number that every configuration of the shape and its fields is assigned, allow it to contain only what the axioms force or the smallest option they leave open, and ask for the configurations at which it does not change. The answer is Einstein's equation with one additional term, and the section

says which axiom is responsible for each part: closure for the curvature and the vacuum term, order for the matter and the force field, growth and decay for the additional term. The equation is twelve-dimensional. Its restriction to the base of K_8 is the system in which Section 4 derives the warp exponent, and its four-dimensional consequences are the subject of later papers of the series; neither is needed to state it. We announce the result.

Theorem 3.1 (the Master Equation)**.** *Under the action* (8) *the metric of* M_{12} *satisfies*

$$\begin{aligned} G_{MN} \;&=\; 8\pi G_{\text{eff}}\, T_{MN} \;+\; \zeta\, T^{\Phi_{\text{QC}}}_{MN} \;+\; \Lambda_{\text{eff}}\, g_{MN}\,, \\ T^{\Phi_{\text{QC}}}_{MN} \;&\equiv\; \langle \nabla_M \Phi_{\text{QC}}, \nabla_N \Phi_{\text{QC}} \rangle_M \,-\, \tfrac{1}{2}\, g_{MN}\, \langle \nabla \Phi_{\text{QC}}, \nabla \Phi_{\text{QC}} \rangle_M\,, \end{aligned} \tag{6}$$

where $\Phi_{\text{QC}} = \omega|\psi - \bar{\psi}|^2$ *is the dilaton of Definition 2.3, a section of the Clifford module* $M \cong \mathbb{H}^{32}$, $\langle u, v \rangle_M = \sum_\alpha \mathrm{Re}(u_\alpha \bar{v}_\alpha)$ *the module inner product,* ζ *the coupling,* T_{MN} *the conserved stress tensor of the spinor sector of A3 together with the field strength of the Spin*^{c} *line,* $8\pi G_{\text{eff}} = \kappa_{12}^2$, *and* Λ_{eff} *collects the vacuum terms. The dilaton satisfies*

$$\zeta\, \Box \Phi_{\text{QC}} \;=\; V'(\Phi_{\text{QC}})\,, \tag{7}$$

so that $\Box \Phi_{\text{QC}} = 0$ *wherever the potential is stationary or absent.*

In words, the tensor $T^{\Phi_{\text{QC}}}_{MN}$ is built as follows. Take the rate of change of the dilaton in direction M and the rate of change in direction N; these are each thirty-two quaternions, and their inner product over the module, summing the real parts of the products component by component, is a single real number $\langle \nabla_M \Phi_{\text{QC}}, \nabla_N \Phi_{\text{QC}} \rangle_M$. That is the first term: the stress the dilaton's variation puts between directions M and N. The second term subtracts half the total squared rate of change of the dilaton, in all directions at once, spread evenly over the metric g_{MN}; it is the pressure the dilaton's gradient exerts isotropically. The difference is the standard stress tensor of a scalar field, familiar from every textbook, with the one change that the squares are inner products over the module. For a single real component it is exactly $\nabla_M \Phi_{\text{QC}} \nabla_N \Phi_{\text{QC}} - \frac{1}{2} g_{MN} |\nabla \Phi_{\text{QC}}|^2$.

Read term by term, (6) is Einstein's equation, $G_{MN} = 8\pi G\, T_{MN} + \Lambda g_{MN}$, with the one additional term $\zeta T^{\Phi_{\text{QC}}}_{MN}$. The curvature on the left and the vacuum term exist because the shape closes on itself (A1); the matter and force term T_{MN} exists because order matters (A3) and the algebra of order acts on spinors; the additional term exists because the shape grows and decays as a function of itself (A2), its field being written on the two sheets of that law; and the two sheets are two, rather than one sheet and its reflection, because the shape never stands still (A4, Section 2.5). Einstein's equation is the case in which the second and fourth axioms are absent.

Two readings are fixed now. By Definition 2.1 the metric is positive-definite and every "time" statement below is a statement along the flow. And ζ in (6) is a coupling constant of the twelve-dimensional theory; Section 4 derives its value, 12π, as the exponent by which the warp of A2 falls across a half-period of the base when the geometry responds to the dilaton's profile. The number is one number seen in two places, and the equation displays it once.

3.1 The action

Definition 3.2 (the action on M_{12})**.**

$$\begin{aligned} S \;=\; \frac{1}{2\kappa_{12}^2} \int_{M_{12}} \sqrt{g}\, \Big[\, R \;-\; 2\Lambda_{12} \;-\; \tfrac{1}{4}\, F_{MN} F^{MN} - \zeta\, \langle \nabla \Phi_{\text{QC}}, \nabla \Phi_{\text{QC}} \rangle_M \;-\; 2V(\Phi_{\text{QC}}) \Big] \\ +\; S_{\text{spinor}}[g, \Psi, A]\,, \end{aligned} \tag{8}$$

with $F = \mathrm{d}A$ the curvature of the induced connection on the determinant line $L = K^{-1}$ (Section 3.1), Φ_{QC} the dilaton of Definition 2.3, V its potential, and S_{spinor} the Dirac action of the Spin^c spinor of A3 on the bundle of M1 Proposition 7.3.

In words: the number S assigned to a configuration is the integral over the whole twelve-dimensional shape of five contributions, all weighted by the volume element $\sqrt{g}$ and divided by twice the gravitational constant κ_{12}^2. The first, R, is the scalar curvature — how much the shape bends at each point. The second, $-2\Lambda_{12}$, is the energy of empty volume, a constant per unit volume. The third, $-\frac{1}{4}F_{MN}F^{MN}$, is the energy of the field strength of the Spin^c line, the one abelian field the geometry carries. The fourth, $-\zeta\langle\nabla\Phi_{\mathrm{QC}},\nabla\Phi_{\mathrm{QC}}\rangle_M$, is the energy stored in the dilaton's variation from point to point, the module inner product of its gradient with itself, multiplied by the coupling ζ. The fifth, $-2V(\Phi_{\mathrm{QC}})$, is the potential energy of the dilaton's value. Added to all of these is the action of the spinor field of the third axiom, the Dirac action S_{spinor}, which depends on the metric, the spinor Ψ and the connection A. Each term is accounted for by an axiom or a reading, and we go through them.

One gravitational term. By A1 the domain is compact without boundary (M1 Definition 2.1), so the action has no boundary terms and its stationary points are Euler–Lagrange equations with no boundary data [12]. The divergence-free symmetric tensors built from the metric and two derivatives in twelve dimensions form the five-term Lovelock family [7]; every term beyond the first carries an independent dimensionful coefficient, an unforced datum under criterion (1) of M1 Definition 3.2, excluded on the ground that excluded the twisted Spin^c lines in M1 Proposition 7.3. The Einstein term survives because without it there is no equation; its coefficient is the declared scale of M1 Theorem 8.1. Once one term linear in second derivatives is admitted its uniqueness holds in every dimension [11, 13, 2]. [D]

The connection of the line. M1 Proposition 7.3 selects the Spin^c structure induced by the almost-complex structure of A3's selected unit, with determinant line $L = K^{-1}$ (M1 Definition 2.5). The connection on L is part of the geometry's own data, and its field strength is the one abelian field strength the geometry carries. Its energy therefore enters (8) inside the gravitational bracket, weighted by $1/2\kappa_{12}^2$: a separate gauge coupling for a connection the geometry already owns would be an unforced datum, excluded by criterion (1). What is *not* fixed by the metric is the flux: the quanta of F through the cycles of K_8 are topological data, with total $N = \chi(K_8)$ by the tadpole condition (Section 4), and they sit on the base cycle because it is the only cycle with cone points at which a localised flux can satisfy the first-order equation of Section 4. [D]

The dilaton term, and why the sheets enter through Φ_{QC} alone. The fundamental fields are the sheets $\psi, \bar{\psi}$ and the Coherent Field ω (Section 2); the spinor's own dynamics are the Dirac action in S_{spinor}. The dilaton $\Phi_{\mathrm{QC}} = \omega|\psi - \bar{\psi}|^2$ is composite and module-valued, so a two-derivative term in it is a particular interaction among the fundamental fields and their gradients, and writing it as the only such term is a statement: the geometry couples to the sheets through Φ_{QC} and through nothing else. Which quadratic form in $\nabla\Phi_{\mathrm{QC}}$ appears is not a further choice: the invariant symmetric forms on an irreducible quaternionic-unitary module form a one-dimensional space, so covariance under the fibre algebra forces the module inner product $\langle\nabla_M\Phi_{\mathrm{QC}},\nabla_N\Phi_{\mathrm{QC}}\rangle_M$ — the trace of the square, not the square of the trace. For a vacuum symmetric under the module the trace is 32 times a single component, and the three readings of "$\zeta\nabla\Phi_{\mathrm{QC}}\nabla\Phi_{\mathrm{QC}}$" for a module-valued field — one component, the trace, the square of the trace — carry the weights 1, 32 and 1024 (`L7_three_readings`); only the second is covariant. [R]We adopt this by the standing principle of Section 2.1, and the reason it is the minimal coupling is Proposition 2.4. If curvature is to respond to the sheets through a scalar that is even under the fold and vanishes both where the sheets coincide and where growth stands against decay, Φ_{QC} is the unique lowest-order candidate, and $|\nabla\Phi_{\mathrm{QC}}|^2$ is then the unique two-derivative term with those properties; the competing terms fail the vanishing conditions ($|\nabla\omega|^2$ does not vanish at $\psi = \bar{\psi}$; $|\nabla\psi|^2$ vanishes nowhere). This is A2 read on the metric: the geometry grows and decays

in response to the asymmetry of the sheets weighted by their coherence, and to nothing else. ζ is the strength of that one coupling and is never absorbed into a redefinition of the field. [D] (minimality)

The potential and the vacuum term. Λ_{12} is the energy of empty twelve-volume; $V(\Phi_{\mathrm{QC}})$ is the dilaton's potential, whose form on the base is obtained in Section 4 from the fibre curvature and Λ_{12} when the action is evaluated on the warped configuration. Neither is a free parameter of this paper: Λ_{12} is fixed by the requirement that the base system of Section 4 admit a solution with the given topology, and V is derived there. Earlier work in the programme took $\Lambda_{12} = 0$ as an axiom of the construction; in this series that statement is the case in which the solvability condition returns zero, to be established or refuted where the fibre radii are derived, and it is not assumed here. [D]

3.2 Variation

Proof of Theorem 3.1. Varying (8) with respect to g^{MN},

$$\begin{aligned} G_{MN} + \Lambda_{12} g_{MN} \;&= \zeta\Big(\langle \nabla_M \Phi_{\mathrm{QC}}, \nabla_N \Phi_{\mathrm{QC}} \rangle_M - \tfrac{1}{2} g_{MN} \langle \nabla \Phi_{\mathrm{QC}}, \nabla \Phi_{\mathrm{QC}} \rangle_M \Big) \;-\; V(\Phi_{\mathrm{QC}})\, g_{MN} \\ &\quad +\; \tfrac{1}{2}\Big(F_{MP} F_N{}^P - \tfrac{1}{4} g_{MN} F^2 \Big) \;+\; \kappa_{12}^2\, T^{\mathrm{sp}}_{MN} \,, \end{aligned} \tag{9}$$

where $T^{\mathrm{sp}}_{MN} = -\frac{2}{\sqrt{g}} \frac{\delta S_{\mathrm{spinor}}}{\delta g^{MN}}$. The first bracket is $T^{\Phi_{\mathrm{QC}}}_{MN}$ with coefficient ζ and nothing else; it is the canonical stress tensor of a module-valued scalar, the trace over M of the scalar form, and its form is forced by the kinetic term, not chosen. For a single component it reduces to $\nabla_M \Phi_{\mathrm{QC}} \nabla_N \Phi_{\mathrm{QC}} - \frac{1}{2} g_{MN} |\nabla \Phi_{\mathrm{QC}}|^2$, which is the display the programme has used; every value of $\langle \Phi_{\mathrm{QC}} \rangle$ in later papers is a trace over the thirty-two components. Writing $8\pi G_{\mathrm{eff}} T_{MN} \equiv \kappa_{12}^2 T^{\mathrm{sp}}_{MN} + \frac{1}{2}(F_{MP} F_N{}^P - \frac{1}{4} g_{MN} F^2)$ and $\Lambda_{\mathrm{eff}} g_{MN} \equiv -\big(\Lambda_{12} + V(\Phi_{\mathrm{QC}})\big) g_{MN}$ gives (6). Varying with respect to Φ_{QC} gives (7). [R] □

The identification of Λ_{eff} carries the dilaton's potential, so it is constant only where V is stationary; this is the reading under which the canonical display and (7) are consistent, as the next proposition shows.

Proposition 3.3 (consistency)**.** $\nabla^M T^{\Phi_{\mathrm{QC}}}_{MN} = \langle \Box \Phi_{\mathrm{QC}}, \nabla_N \Phi_{\mathrm{QC}} \rangle_M$. *Hence the contracted Bianchi identity, the conservation of* T_{MN} *and* (6) *imply*

$$\zeta \langle \Box \Phi_{\mathrm{QC}}, \nabla_N \Phi_{\mathrm{QC}} \rangle_M = \nabla_N V = \langle V'(\Phi_{\mathrm{QC}}), \nabla_N \Phi_{\mathrm{QC}} \rangle_M \,,$$

which is (7) *on the support of* $\nabla \Phi_{\mathrm{QC}}$*; and where* $V' = 0$, $\Box \Phi_{\mathrm{QC}} = 0$ *there. The computation is componentwise and the trace is linear.*

Proof. Take the divergence of the stress tensor, that is, contract its first index with a derivative. The first term, the inner product of two gradients, gives by the product rule two pieces: the Laplacian of the dilaton paired with its gradient, $\langle \Box \Phi_{\mathrm{QC}}, \nabla_N \Phi_{\mathrm{QC}} \rangle_M$, and the gradient paired with the second derivative, $\langle \nabla^M \Phi_{\mathrm{QC}}, \nabla_M \nabla_N \Phi_{\mathrm{QC}} \rangle_M$. The second term, half the squared gradient spread over the metric, gives on differentiation $\langle \nabla^M \Phi_{\mathrm{QC}}, \nabla_N \nabla_M \Phi_{\mathrm{QC}} \rangle_M$, the gradient paired with the second derivative in the other order. For a scalar the two orders of covariant differentiation agree, so those two pieces cancel and only $\langle \Box \Phi_{\mathrm{QC}}, \nabla_N \Phi_{\mathrm{QC}} \rangle_M$ survives:

$$\nabla^M \big(\langle \nabla_M \Phi_{\mathrm{QC}}, \nabla_N \Phi_{\mathrm{QC}} \rangle_M \big) - \tfrac{1}{2} \nabla_N \langle \nabla \Phi_{\mathrm{QC}}, \nabla \Phi_{\mathrm{QC}} \rangle_M = \langle \Box \Phi_{\mathrm{QC}}, \nabla_N \Phi_{\mathrm{QC}} \rangle_M \,.$$

Now take the divergence of both sides of (6). The left side vanishes by the contracted Bianchi identity, the geometric fact that the Einstein tensor is automatically divergence-free. The matter term vanishes because T_{MN} is conserved (Section 3.4). The vacuum term contributes the

gradient of Λ_{eff}, which is the gradient of $-V(\Phi_{\text{QC}})$, that is $-\langle V'(\Phi_{\text{QC}}), \nabla_N \Phi_{\text{QC}} \rangle_M$. What is left is

$$\zeta \langle \Box \Phi_{\text{QC}}, \nabla_N \Phi_{\text{QC}} \rangle_M = \langle V'(\Phi_{\text{QC}}), \nabla_N \Phi_{\text{QC}} \rangle_M \,,$$

which is (7) wherever the dilaton is actually varying; and where the potential is flat, $V' = 0$, it says the dilaton satisfies the wave equation $\Box \Phi_{\text{QC}} = 0$ there. The computation is componentwise and the trace is linear, so nothing changes in passing from one component to the module. [R] □

Proposition 3.4 (the dilaton stress tensor along the flow)**.** *Let Φ_{QC} depend only on the flow parameter s of A4. In an orthonormal frame $(\partial_s, e_1, \ldots, e_{11})$ the tensor $\zeta T^{\Phi_{\text{QC}}}$ is diagonal with*

$$\zeta T^{\Phi_{\text{QC}}}_{ss} = \tfrac{1}{2} \zeta \langle \dot{\Phi}_{\text{QC}}, \dot{\Phi}_{\text{QC}} \rangle_M \,, \qquad \zeta T^{\Phi_{\text{QC}}}_{ii} = -\tfrac{1}{2} \zeta \langle \dot{\Phi}_{\text{QC}}, \dot{\Phi}_{\text{QC}} \rangle_M \quad (i = 1, \ldots, 11) \,. \tag{10}$$

Proof. If the dilaton depends only on the flow parameter, its gradient points along the flow and nowhere else: $\nabla \Phi_{\text{QC}} = \dot{\Phi}_{\text{QC}}\, \partial_s$, where the dot is the derivative along the flow. Its squared length is then just $\langle \dot{\Phi}_{\text{QC}}, \dot{\Phi}_{\text{QC}} \rangle_M$. In the component along the flow the first term of the stress tensor contributes the whole of that, and the second term subtracts half of it, leaving a half. In any direction transverse to the flow the first term contributes nothing, because the gradient has no component there, and the second term subtracts half the squared length, leaving minus a half. Off-diagonal components vanish for the same reason (`L2_flow_components`). [R] □

The reading. In a Lorentzian presentation, with the flow parameter continued to a signed time, the two entries of (10) are an energy density and an isotropic pressure with $\rho_{\Phi_{\text{QC}}} = p_{\Phi_{\text{QC}}} = \tfrac{1}{2} \zeta \langle \dot{\Phi}_{\text{QC}}, \dot{\Phi}_{\text{QC}} \rangle_M$, the stiff equation of state, and (7) with $V' = 0$ gives $\rho_{\Phi_{\text{QC}}} \propto a^{-6}$ on an isotropic background. These are the statements the programme uses. Under Definition 2.1 they are the components (10) read along the flow, the transverse sign carried by the continuation; the equation is (6) in either reading. [D]

3.3 The base restriction

Section 4 works on the base $T^2/\mathbb{Z}_2$ with the fibre warped. We record here, so that Section 4 may cite it, what the Master Equation becomes there. Evaluate (8) on the configuration of Definition 2.5: metric $g_4 + e^{2a(\theta)} \hat{g}_6 + g_2(\theta)$, dilaton $\Phi_{\text{QC}} = \Phi_{\text{QC}}(\theta)$, the connection induced. The fibre volume contributes e^{6a}, its curvature $e^{-2a} \hat{R}_6$, and the warped-product scalar curvature contributes the gradient terms of a [8, Ch. 7]; the result is a two-dimensional action on the base of the form

$$S_2 \;=\; \int_{T^2/\mathbb{Z}_2} \sqrt{g_2} \Big[\frac{1}{2\kappa_2^2} R_2 \;-\; \tfrac{1}{2} |F|^2 \;-\; \mathcal{V}(a, \Phi_{\text{QC}}) \;-\; \mathcal{K}(a, \nabla a, \nabla \Phi_{\text{QC}}) \Big] \,, \tag{11}$$

with κ_2^{-2} the fibre-volume-weighted gravitational coefficient, $\mathcal{V}$ the potential assembled from Λ_{12}, $\hat{R}_6$ and $V(\Phi_{\text{QC}})$ with the weights e^{6a} and e^{4a}, and $\mathcal{K}$ the gradient terms of a and Φ_{QC}. The warp of A2 is the geometry's response to the dilaton: on the solutions of Section 4 the two profiles are tied by

$$a \;=\; -\zeta\, \Phi_{\text{QC}} \,, \tag{12}$$

with Φ_{QC} falling by one unit across a half-period, so that the warp exponent across the half-period is ζ and (11) collapses to the single-modulus Einstein–Maxwell system with potential that Section 4 solves. That system is therefore the base restriction of (6), not an axiom. [D]

3.4 The matter sector

T_{MN} in (6) is the stress tensor of the Spin^c Dirac field of A3 on the bundle with $L = \mathcal{O}(3)$ (M1 Proposition 7.3) with the Clifford action of the fibre algebra (M1 Definition 2.5), together with the field strength of the induced connection. The one property of T_{MN} used in this paper is its

conservation, which follows from the diffeomorphism invariance of S_{spinor} on the closed manifold M_{12} (A1) by the standard argument [12]; for the field strength it is the Bianchi identity $\mathrm{d}F = 0$ and the Maxwell equation. The content of T_{MN}, its decomposition into generations and its masses, belongs to later papers of the series and is not opened here. [D]

3.5 What is derived and what is read

The form of (6) is a theorem of the action (8) ([R]). The action is fixed by one derivation and three readings: the single gravitational term (minimality, [D]), the induced connection ([D]), the coupling of the geometry to the sheets through Φ_{QC} alone ([D], with Proposition 2.4 as its reason), and the Riemannian metric with time along the flow (Definition 2.1). The coupling ζ is a constant of the theory whose value is derived in Section 4; the relation (12) between the warp and the dilaton is a property of the solutions established there.

4 The warp exponent and the value of ζ

The purpose of this section is to solve the base restriction of the Master Equation, (11), for the profile of the warp across a half-period of the pillowcase, and to show that the exponent by which the warp falls is $N\pi$ with N the number of flux units, that $N = \chi(K_8) = 12$, and hence that the coupling of Section 3 has the value $\zeta = 12\pi$. We announce the two results.

Theorem 4.1 (exact warp exponent per flux unit)**.** *For the base system* (11) *on* $T^2/\mathbb{Z}_2$ *with* N *units of quantised flux localised at a cone point, the warp exponent across the half-period transverse to the wall on which the flux sits satisfies exactly*

$$kL \;=\; N\pi\,, \tag{13}$$

where k *is the constant decay rate of the warp exponent between the walls and* L *the corresponding half-period. The factor* π *is the deficit angle at the cone point.*

Corollary 4.2 (the value of ζ)**.** *With* $N = \chi(K_8) = 12$ *(Proposition 4.5) and the dilaton normalised to fall by one unit across the half-period (*(12)*),*

$$\zeta \;=\; kL \;=\; \chi(K_8)\,\pi \;=\; 12\pi\,. \tag{14}$$

4.1 The topological integers

Proposition 4.3. $\chi(T^2/\mathbb{Z}_2) = 2$*, and the total curvature of the pillowcase is* 4π*, concentrated as a deficit of exactly* π *at each of the four cone points.*

Proof. We give two independent routes to the Euler characteristic, and then the curvature. First route, by the cone points. The Euler characteristic of a torus is zero. Folding it in half by the map $z \mapsto -z$ would halve that to zero again, except at the four points the fold leaves fixed, where the quotient is not smooth: each fixed point contributes a correction of $1 - \frac{1}{2} = \frac{1}{2}$, because a neighbourhood of it is a cone of angle π, half a full turn. Four corrections of a half make two, so $\chi = 0 + 4 \cdot \frac{1}{2} = 2$ (M1 Theorem 5.7, `L2_chi_pillowcase`). Second route, by counting holes. The folded torus is, as a topological space, a sphere — stretch the pillowcase out and it is a ball with four puckered points — and a sphere has one connected piece, no independent loops and one enclosed cavity, Betti numbers $(1, 0, 1)$, so $\chi = 1 - 0 + 1 = 2$. The two routes agree. For the curvature, the Gauss–Bonnet theorem for orbifolds [5] says the total curvature is 2π times the Euler characteristic, 4π. The pillowcase is flat everywhere except at the four cone points, so all 4π of curvature sits at them, π at each — which is exactly the angle missing at a cone of angle π (`L5_deficit`). [R] □

Proposition 4.4. $\chi(K_8) = \chi(\mathbb{CP}^2)\,\chi(S^2)\,\chi(T^2/\mathbb{Z}_2) = 3 \cdot 2 \cdot 2 = 12$.

Proof. Multiplicativity of χ over fibrations with compact fibre; the warp does not change the additive structure of cohomology (M1 Theorem 7.6, `L3_chi_K8`). [R] □

Proposition 4.5 (the flux number)**.** *Tadpole cancellation on K_8 fixes the number of quantised flux units of the determinant line to $N_{\text{flux}} = \chi(K_8) = 12$, and all twelve sit on the base cycle.*

Proof. On a compact space the total charge sourced must cancel. The gravitational contribution to the tadpole is the Euler density, whose integral is $\chi(K_8)$; the flux contribution is integrally quantised and must balance it, so $N_{\text{flux}} = \chi(K_8) = 12$. Cross-check from the parity structure: the individual quanta are odd on the non-spin factor $\mathbb{CP}^2$ and even on the spin factor S^2, by $c_1 \equiv w_2 \bmod 2$ (M1 Proposition 7.3, `L4_flux_parity`), and the values $(Q_c, Q_w) = (3, 6)$ of M1 Proposition 8.2 are the ones the constraint admits. The twelve units sit on the base because a localised flux satisfying the first-order equation of Section 4.3 needs a cone point to sit at, and the base is the only cycle that has them. The smallest case: one unit at one corner balanced by one unit of integrated Euler density on its cone neighbourhood; the general case is the sum of twelve such balances. [D] □

4.2 The base system and its single modulus

Section 3.3 obtained the base restriction (11) of the Master Equation: two-dimensional gravity with coefficient κ_2^{-2}, the energy $\frac{1}{2}|F|^2$ of the line's field strength, a potential, and the gradient terms of the fibre warp a and the dilaton Φ_{QC}. Two facts reduce it to a system with a single modulus.

The fibre decouples from the jump. $\mathbb{CP}^2$ with the Fubini–Study metric and S^2 with the round metric are Einstein, $\text{Ric} = (6/R_c^2)g$ and $\text{Ric} = (1/R_w^2)g$ (M1 Propositions 7.1, 7.2). For a warp depending only on the base coordinates the warped-product curvature [1] therefore contributes to the base equation only smooth scalar terms in a — a potential and a modification of the gradient term — with no mixed base–fibre components. Smooth terms cannot change a jump condition at a cone point; they change the shape of the profile between cone points. The effective two-dimensional gravitational coefficient is the fibre-volume-weighted one,

$$\kappa_2^2 \;=\; \frac{\kappa_{12}^2}{V_4\,\text{Vol}_0\,e^{6a_0}}\,, \qquad \text{Vol}_0 = \text{Vol}(\mathbb{CP}^2)\,\text{Vol}(S^2)\,, \tag{15}$$

with V_4 the external volume and a_0 the warp at the wall. [D]

The dilaton and the warp are one modulus. On the solutions constructed below the dilaton's profile and the fibre warp are tied by (12), $a = -\zeta\Phi_{\text{QC}}$, with Φ_{QC} the canonically normalised modulus of the warped direction and $\Phi_{\text{QC}} = 1$ the configuration at the far wall. This is the definition of the normalisation of Φ_{QC} on the base, and it is what makes the coupling ζ of (6) and the exponent of (13) one number: a profile of Φ_{QC} that falls by one unit across the half-period is a warp exponent that falls by ζ. In the fossil form of this construction the transverse base length itself was written with the factor $e^{2\zeta\Phi_{\text{QC}}}$ while the fibre carried e^{2a}; under (12) these are two expressions of the same exponent and not two warps. [D]

With these two facts (11) is the single-modulus system

$$S_2 \;=\; \int_{T^2/\mathbb{Z}_2} \sqrt{g_2}\,\Big[\frac{1}{2\kappa_2^2}R_2 \;-\; \tfrac{1}{2}|F|^2 \;-\; V(a)\Big]\,, \tag{16}$$

with $V(a)$ the bulk potential assembled from Λ_{12}, the fibre curvature and the dilaton potential, continuous on the base. It is the restriction of the Master Equation and not an axiom. [D]

4.3 Bogomol'nyi completion

Let θ be the base coordinate transverse to the wall on which the flux sits, and consider a static profile $a(\theta)$. The energy functional of (16) admits a completion of the square once $V(a)$ balances the geometric bulk terms,

$$\mathcal{E} \;\propto\; \int \sqrt{g_2}\,\Big(\partial_\theta a \;\pm\; \frac{\kappa_2^2}{2}\,\frac{|F_{\psi\omega}|}{\sqrt{g_2}}\Big)^2 \;\geq\; 0\,, \tag{17}$$

saturated when the square vanishes. The balance of V against the bulk terms is the familiar one of warped five-dimensional models [9]; here it is not a tuning but the condition that Λ_{12} be the value at which the base system admits a solution with the given topology, which is how Λ_{12} is fixed in this series (Section 3.1). The negative root, a decaying warp, gives the first-order equation

$$\partial_\theta a \;=\; -\frac{\kappa_2^2}{2}\,\frac{|F_{\psi\omega}|}{\sqrt{g_2}}\,, \tag{18}$$

and everything below is an exact consequence of it. [R] (as a theorem of (16)); the balance condition is [D].

4.4 Flux localisation and quantisation

Dirac quantisation over the orbifold is $\frac{1}{2\pi}\int F = N$. For N units at the drive-fixed corner $P_1 = (0,0)$ — the one corner fixed by both the fold and the drive (M1 Propositions 4.3, 5.6) — the density is

$$F_{\psi\omega} \;=\; N\pi\,\frac{\delta(\theta_\psi)\,\delta(\theta_\omega)}{\sqrt{g_2}}\,, \tag{19}$$

checked directly: over the fundamental domain $\frac{1}{2\pi}\int F = N/2$, and the domain is half of the covering torus, so the total over the cover is $2\cdot N/2 = N$ (`L5_FD_check`). [R]

4.5 Solving the first-order equation

Away from the corner the localised flux vanishes and the flow is driven by the constant bulk potential, giving the linear profile

$$\partial_\theta a = -k \qquad\Longrightarrow\qquad a(\theta) = a(0) - k|\theta|\,; \tag{20}$$

that the slope is constant between the walls is the boundary condition of the construction, and the fibre's smooth backreaction, absorbed in V, alters the shape between the walls without touching what follows. At the corner, integrating (18) across $[-\epsilon,\epsilon]$,

$$[\partial_\theta a]_0 \;=\; -\frac{\kappa_2^2}{2}\int_{-\epsilon}^{\epsilon}\frac{|F_{\psi\omega}|}{\sqrt{g_2}}\,\mathrm{d}\theta \;=\; -\kappa_2^2 N\pi\,, \tag{21}$$

exactly: each unit of flux contributes a kink of exactly $\kappa_2^2\pi$, and no smooth term contributes as $\epsilon\to 0$. From (20) the same kink is $a'(0^+) - a'(0^-) = -2k$, so

$$2k \;=\; \kappa_2^2\,N\pi\,. \tag{22}$$

[R] (the jump); [D] (constant slope).

4.6 Gauss–Bonnet normalisation

The half-period L and the two-dimensional coupling are not independent. By Proposition 4.3 the total curvature 4π sits as π at each corner; the normalisation connecting the deficit angles to the length of the domain fixes

$$\kappa_2^2 \;=\; \frac{2}{L}\,, \tag{23}$$

which, with (15), is a relation among the scales of the geometry: $\kappa_{12}^2 = 2V_4\mathrm{Vol}_0 e^{6a_0}/L$, the half-period measured in the gravitational unit weighted by the fibre volume. It is the one normalisation of this section and we tag it as such. [D]

4.7 The result

Substituting (23) in (22), $2k = (2/L)N\pi$, hence $kL = N\pi$, which is Theorem 4.1 (`L5_balance`). The smallest case by hand: $N = 1$, one unit at one corner, kink $\kappa_2^2\pi = 2\pi/L$, rate $k = \pi/L$, exponent across the half-period exactly π; the general case is N copies of this balance. Corollary 4.2 is the composition with Proposition 4.5: $\zeta = kL = \chi(K_8)\pi = 12\pi$ (`L5_zeta`). Three checks: (i) if the fundamental domain degenerates to $S^1/\mathbb{Z}_2$ the two-form flux becomes a brane tension and (21) is the Israel junction condition of [9]; (ii) the saturated configuration has positive-definite flux energy; (iii) $\partial_\theta a = -k < 0$ between the walls, so the warp decreases monotonically. [R]

4.8 Which direction, and the two sheets

Nothing above depends on whether the transverse coordinate θ is θ_ψ or θ_ω: the normalisation (23) is applied per direction and (13) is direction-blind. The sheet exchange τ of Section 2.5 fixes P_1 and P_4 and exchanges P_2 and P_3, the ψ- and ω-half-periods (`L5_tau_corners`), so a profile transverse to the wall of one sheet is, under τ, a profile transverse to the wall of the other. The exponent 12π is therefore τ-symmetric, and calling the transverse period L_ω or L_ψ is the choice of which sheet one calls ψ, the convention of M1 Appendix B. What is not conventional is the aspect ratio: the two half-periods differ by $\tau_2 = L_\omega/L_\psi = 1.8466$ (M1 Proposition 8.2), so the rate k differs by τ_2 between the sheets, and any quantity built from a period length rather than from the exponent depends on which sheet is described. In this series "$L_\omega = 12\pi$" means the half-period transverse to the ψ-wall, in the sheet called ψ, in the unit of (23). [D]

4.9 What is derived and what is read

The jump (21), the balance (22), the integers of Section 4.1 and the composition (14) are theorems ([R]). Read, and named: the placement of the twelve units on the base cycle ([D], Proposition 4.5); the constant slope between walls ([D], boundary condition); the normalisation (23) ([D]); the tie $a = -\zeta\Phi_{\mathrm{QC}}$ with $\Phi_{\mathrm{QC}} = 1$ at the far wall ([D], the definition of the dilaton's normalisation); and the balance of V as the determination of Λ_{12} ([D]). The value $\zeta = 12\pi$ rests on these and on nothing outside this paper and M1.

5 What the Coherent Field is

The Coherent Field entered in Definition 2.3 as the even combination of the sheets. This section says what it measures. The programme reads ω as a one-hump quantity: zero when the sheets are rigidly locked, zero when they are uncorrelated, positive between. We derive the two zeros from A2 and A4, show where the maximum of the coupling sits, settle the sign of ω on the solutions of Section 4, and name what this paper does not prove.

5.1 The phase of a sheet field

Work in the corner reading of Definition 2.2, where each component is complex, and write $\psi = re^{i\vartheta}$ in $\mathbb{C}_u$, so that $\omega = 2r\cos\vartheta$ and $\psi - \bar{\psi} = 2ir\sin\vartheta$; the statements below hold componentwise, and the module trace sums them. The phase ϑ is the angle between the two sheets: $\vartheta = 0$ when they coincide, $\vartheta = \pm\pi/2$ when growth stands exactly against decay. By Definition 2.2, $\hat{\tau}$ sends $\vartheta \mapsto -\vartheta$.

Proposition 5.1 (the locked zero)**.** *$\omega = 0$ at a point iff $\bar{\psi} = -\psi$ there, i.e. iff the sheet field is $\hat{\tau}$-odd, $\vartheta = \pm\pi/2$. At such a point the dilaton vanishes, $\Phi_{\mathrm{QC}} = 0$, and the geometry does not respond to the sheets.*

Proof. The Coherent Field is ψ plus its shadow. It is zero at a point exactly when the shadow there is the negative of the field, $\hat{\tau}\psi = -\psi$: when exchanging growth and decay reverses the sign of every component. In the corner reading, where a component is $re^{i\vartheta}$ and the exchange sends ϑ to $-\vartheta$, this says $re^{i\vartheta} = -re^{-i\vartheta}$, that is $\cos\vartheta = 0$, so $\vartheta = \pm\pi/2$: the two sheets stand at right angles, growth exactly against decay. At such a point the dilaton $\Phi_{\mathrm{QC}} = \omega|\psi - \bar{\psi}|^2$ is zero because its first factor is, and so the ζ-term of (6) vanishes there: the geometry does not respond. [R] □

This is A2 read on the field: when growth is exactly opposed by decay the two sheets are locked, nothing in the geometry changes, and the even combination is zero. It is the reason the Coherent Field is the combination $\psi + \bar{\psi}$ and not the modulus, which does not vanish there (Section 2.3). We call this the *locked* or rigid configuration.

5.2 The uncorrelated zero, from the drive

The second zero comes from A4. The drive M_0 acts on the base torus (M1 Theorem 4.2), and a sheet field is carried along its orbit, $\psi \mapsto \psi \circ M_0^n$. We make precise the average that the programme's "orbit average" refers to.

Definition 5.2 (orbit average)**.** For a sheet field ψ and a point z of the base, the orbit average of ω over T circuits is $\langle\omega\rangle_T(z) = \frac{1}{T}\sum_{n<T}\omega(M_0^n z)$. The *phase distribution* of ψ is the push-forward of Lebesgue measure on the base under $z \mapsto \vartheta(z)$.

Proposition 5.3 (the uncorrelated zero)**.** *The drive is ergodic for Lebesgue measure, so for almost every z, $\langle\omega\rangle_T(z) \to \int \omega \,\mathrm{d}\mu$ as $T \to \infty$: the orbit average is the space average. If the phase distribution of ψ is uniform on the circle — the phase of the sheets is uncorrelated with position — then $\int \omega \,\mathrm{d}\mu = 0$.*

Proof. A hyperbolic toral automorphism is ergodic, indeed mixing, for Lebesgue measure [4]; M1 Proposition 4.1 shows A4 forces hyperbolicity. Birkhoff's theorem gives the first statement. For the second, with r bounded and ϑ uniform, $\int\omega = 2\int r\cos\vartheta\,\mathrm{d}\mu = 2\bar{r}\int_0^{2\pi}\cos\vartheta\,\frac{\mathrm{d}\vartheta}{2\pi} = 0$, where the factorisation uses that a uniform phase is independent of the modulus. [R] (given the independence, which is what "uniform phase" means). □

This is A4 read on the field: a drive of positive entropy mixes, and a sheet field whose phase has been mixed to uniformity has no net even combination. We call this the *uncorrelated* configuration.

5.3 The hump, and where the coupling is largest

Between the two zeros the average is positive. Let the phase distribution be concentrated around a lock angle ϑ_0 with concentration $c \geq 0$ (any unimodal family will do; the circular

normal $\propto e^{c\cos(\vartheta-\vartheta_0)}$ is the standard one). Then

$$\langle\omega\rangle \;=\; 2\bar{r}\,A(c)\cos\vartheta_0\,, \qquad A(c)=\frac{I_1(c)}{I_0(c)}\in[0,1)\,, \tag{24}$$

which vanishes at $c=0$ (uniform, Proposition 5.3), vanishes at $\vartheta_0=\pm\pi/2$ (locked, Proposition 5.1), and is positive for $|\vartheta_0|<\pi/2$: one hump in the lock angle between the two locked configurations. [D]

The coupling to curvature is not maximal where ω is. In the phase,

$$\Phi_{\mathrm{QC}} \;=\; \omega\,|\psi-\bar{\psi}|^2 \;=\; 8r^3\cos\vartheta\,\sin^2\vartheta\,, \tag{25}$$

which vanishes at $\vartheta=0$ (sheets coincident, no held distinction) and at $\vartheta=\pm\pi/2$ (locked), and is extremal where $\tan^2\vartheta=2$, $\sin^2\vartheta=\frac{2}{3}$. The geometry responds most strongly neither when the sheets agree nor when they oppose, but at the definite intermediate angle $\vartheta^*=\arctan\sqrt{2}\approx 54.7^\circ$ (`L6_Phi_extremum`). [R]

5.4 The sign of ω on the solutions

Section 4 gave the warp as a decaying exponential, $a=-\zeta\Phi_{\mathrm{QC}}$ with Φ_{QC} rising from 0 at the wall to 1 at the far wall, Φ_{QC} there standing for the real part of the trace. On those solutions, therefore, $\Phi_{\mathrm{QC}}\geq 0$. But Φ_{QC} is ω times a non-negative real number, $|\psi-\bar{\psi}|^2$, so wherever the sheets are distinct — wherever that number is not zero — the real part of ω has the same sign as Φ_{QC} and is non-negative: in the corner reading, $\cos\vartheta\geq 0$, which is the half-circle $|\vartheta|\leq\pi/2$. The Coherent Field does not change sign on the solutions. What fixed the sign was the direction of the warp: the exponential decays away from the wall, so Φ_{QC} rises. Had the warp grown instead, the half-circle would be the other one, and the overall sign of ψ — which no axiom fixes, since $\psi\mapsto-\psi$ commutes with $\hat{\tau}$ and reverses ω and Φ_{QC} together — would be chosen to restore $\mathrm{Re}\,\omega\geq 0$. Away from the corner reading ω carries an imaginary part along the selected unit, which the sign statement does not constrain. [D]

5.5 What this paper does not prove

The programme's canonical reading of ω is a one-hump complexity functional of a *spectrum*: zero for a rigidly ordered spectrum, zero for an uncorrelated one, maximal in the regime of strong level repulsion, with spectral rigidity as the computable proxy. The two zeros derived above are the field-level counterparts of the first two statements: locked for rigidly ordered, uniform for uncorrelated. The identification of the field ω with that spectral functional, and the placement of its maximum in the regime of strong level repulsion, are not proved here; they require the spectrum of the Dirac operator at the cone points of the pillowcase and a bridge theorem between the two descriptions, which is the subject of a later paper of the series. We record the reading as motivated and the bridge as open. [M]

6 Applications and outlook

The equation (6) with $\zeta=12\pi$ is the one the papers of the programme have used, and this paper fixes its origin: the form from the variation, the coupling from the warp exponent, the variables from the sheets. By axiom: closure gives the equation and the vacuum term, order gives matter and the forces, growth and decay gives the additional term, the drive gives the arrow. Three things follow for the physics without further input. The two sectors ψ and $\bar{\psi}$ are distinct by handedness relative to the flow, so the shadow sector of the programme is not a relabelling. The stress tensor of the dilaton along the flow is stiff, so wherever Φ_{QC} is driven the geometry responds as the programme's collapse and budget analyses assume, read along the flow. And the

warp falls by exactly 12π across a half-period of the base, which is the number from which the programme's hierarchies descend. The papers that follow derive from the same second axiom, through the walls of the sheets and the phase of the dilaton, the masses, the three generations and the mixing angles of the quarks and leptons and the Higgs sector: the structure of matter is the structure of growth and decay, and what a theory of four forces left unexplained, why there is matter and why it does not run backwards, is supplied by the two axioms such a theory omits. We do not pursue those derivations here. [M]

Four things are left open and named. The spectral face of ω, its identification with a one-hump functional of the Dirac spectrum at the cone points, is the subject of a later paper (Section 5.5). The normalisation $\kappa_2^2 = 2/L$ is reproduced here as a relation among the scales and tagged [D]; a derivation from the orbifold Gauss–Bonnet theorem with the warped weights would complete it (Section 4.6). The content of the matter stress tensor, its generations and masses, belongs to the papers that follow (Section 3.4). And the ratio of the drive's rate per circuit, $\log \varphi^2$, to the warp's exponent per half-period, 12π, is a pure number of the geometry whose meaning this paper does not determine (Section 2.5).

Acknowledgements

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A Machine-verified certificates

Seven certificates accompany the paper, each a single Lean 4.15.0 file on the core toolchain with no Mathlib dependency, no author axioms and no `sorry`. They build within the series' proof tree `MEFProof.Paper_M2`; the build record is

```
Built MEFProof.Paper_M2.M2_L1_parity      (1.2s)
Built MEFProof.Paper_M2.M2_L2_master      (549ms)
Built MEFProof.Paper_M2.M2_L3_zeta        (626ms)
Built MEFProof.Paper_M2.M2_L4_handedness  (1.2s)
Built MEFProof.Paper_M2.M2_L5_warp        (649ms)
Built MEFProof.Paper_M2.M2_L6_omega       (608ms)
Built MEFProof.Paper_M2.M2_L7_module      (863ms)
```

Every certified statement is named at the point in the text where it is used. The analytic steps — the variation, the Bianchi contraction, the ergodic theorem, the Pin anticommutation — are one-line computations or cited theorems and are not certified; the certificates fix the arithmetic and the finite case analyses those steps rely on.

`M2_L1_parity.lean` (Section 2, 12 theorems): the $\hat{\tau}$-sign of a monomial $(\mathrm{Re}\,\psi)^a(\mathrm{Im}\,\psi)^b$ is $(-1)^b$; vanishing on $\psi = \bar{\psi}$ iff $b \geq 1$ and on $\psi = -\bar{\psi}$ iff $a \geq 1$; degree bound $a + b \geq 3$ and uniqueness of $(1, 2)$; the parity table of ω, $\psi - \bar{\psi}$, $|\psi - \bar{\psi}|^2$, Φ_{QC}; multiplicativity of the sign.

`M2_L2_master.lean` (Section 3, 6): the flow components of $T^{\Phi_{\mathrm{QC}}}$ and the trace in twelve dimensions; $|\nabla\Phi_{\mathrm{QC}}|^2$ is $\hat{\tau}$-even of degree six; the competing two-derivative terms fail the vanishing conditions.

`M2_L3_zeta.lean` (Section 4, 7): $\chi(K_8) = 3 \cdot 2 \cdot 2$; deficit π per cone, total $4\pi = 2\pi\chi$; $\zeta/\pi = 12$ and the identity of the cone-deficit and Euler readings.

`M2_L4_handedness.lean` (Section 2.5, 11): $SM_0S^{-1} = M_0^{-1}$; $S^2 = -I$; $\det S = 1$ and the orientation of τ; the eigen-directions $(\varphi, 1)$, $(-1, \varphi)$ in $\mathbb{Z}[\varphi]$ with eigenvalues $\varphi^{\pm 2}$; S exchanges them; any involutive flow reversal has determinant -1.

`M2_L5_warp.lean` (Section 4, 9): the orbifold Euler count; the fundamental-domain check; the balance $kL = N\pi$; $\zeta = 12\pi$ and the $N = 1$ case; τ fixes P_1, P_4 and exchanges P_2, P_3.

`M2_L6_omega.lean` (Section 5, 5): the coefficient 8 in $\Phi_{\mathrm{QC}} = 8r^3 \cos\vartheta \sin^2\vartheta$; the extremum $\tan^2\vartheta = 2$; the sign bookkeeping of the half-circle; the zeros of the hump.

`M2_L7_module.lean` (Sections 2, 3, 8): $\mathrm{Cl}(0,12) \cong \mathbb{H}(32)$ by Bott periodicity ($12 \equiv 4 \bmod 8$) and the module dimensions 32, 64, 128; the three readings of the kinetic term, 1, 32, 1024; the trace of a module-symmetric vacuum; the square of an imaginary quaternion is minus its squared modulus.

The seven files follow verbatim.

`M2_L1_parity.lean` — the parity algebra

```
/-!
# M2 — Certificate L1: the parity algebra of ψ, ω and Φ_QC (Section 2 / WP2)

Core toolchain, Lean 4.15.0, no Mathlib, no author axioms, zero `sorry`.

Model. The field ψ has real part x = Re ψ and imaginary part y = Im ψ. The
involution ι of A2 acts on ψ by conjugation (M1 Prop 5.4/5.6), so on (x, y)
it acts as (x, −y). A real monomial in the two components is (x^a)(y^b)
with natural exponents (a, b). Under ι it picks up the sign −(1)^b.

Certified:
* `L1_iota_sign`      — the ι-sign of (a, b) is +1 iff b is even;
* `L1_vanish_Im`      — (x^a)(y^b) vanishes on the locus y = 0 ψ( = ψ⁻) for all x
                        iff b ≥ 1;
* `L1_vanish_Re`      — (x^a)(y^b) vanishes on the locus x = 0 ψ( = −ψ⁻, ω = 0)
                        for all y iff a ≥ 1;
* `L1_degree_bound`   — any ι-even monomial vanishing on both loci has degree ≥ 3;
* `L1_degree_three`   — degree 3 forces (a, b) = (1, 2), i.e. (Re ψ)(Im ψ)²,
                        the monomial of ωψ( − ψ⁻)² up to a real constant;
* `L1_table_*`        — the parity table: ω = 2 Re ψ even; ψ − ψ⁻= 2i Im ψ odd;
                        ψ( − ψ⁻)² even; Φ_QC = ωψ( − ψ⁻)² even.

The two vanishing conditions are inputs of the theorem, not outputs: ω = 0 atψ
 = −ψ⁻is the PI ruling R1 (M2 §2); Φ_QC = 0 at ψ = ψ⁻is the motivated
condition of M2 §2 (no held distinction, no coupling). The certificate shows
that, given both, parity and minimal degree determine the monomial uniquely.
-/

namespace M2L1

/-- The ι-sign of the monomial x^a y^b: +1 if b even, −1 if b odd. -/
def iotaSign (b : Nat) : Int := if b % 2 = 0 then 1 else -1

theorem L1_iota_sign (b : Nat) : iotaSign b = 1 ↔ b % 2 = 0 := by
  unfold iotaSign
  split
  · simp_all
  · constructor
    · intro h; cases h
    · intro h; contradiction

/-- Evaluation of the monomial over ℕ (vanishing is a statement about zeros). -/
def ev (a b x y : Nat) : Nat := x ^ a * y ^ b

theorem zero_pow_of_pos {n : Nat} (h : 0 < n) : (0:Nat) ^ n = 0 := by
  cases n with
  | zero => cases h
  | succ k => simp [Nat.pow_succ]

/-- x^a y^b vanishes at y = 0 for every x iff b ≥ 1. -/
theorem L1_vanish_Im (a b : Nat) : ∀( x, ev a b x 0 = 0) ↔ 1 ≤ b := by
```

```
  constructor
  · intro h
    cases b with
    | zero => have := h 1; simp [ev] at this
    | succ k => omega
  · intro hb x
    simp [ev, zero_pow_of_pos hb]

/-- x^a y^b vanishes at x = 0 for every y iff a ≥ 1. -/
theorem L1_vanish_Re (a b : Nat) : ∀( y, ev a b 0 y = 0) ↔ 1 ≤ a := by
  constructor
  · intro h
    cases a with
    | zero => have := h 1; simp [ev] at this
    | succ k => omega
  · intro ha y
    simp [ev, zero_pow_of_pos ha]

/-- ι-even and vanishing on both loci forces degree a + b ≥ 3. -/
theorem L1_degree_bound (a b : Nat)
    (heven : b % 2 = 0) (ha : 1 ≤ a) (hb : 1 ≤ b) : 3 ≤ a + b := by
  omega

/-- At the minimal degree 3 the exponents are (1, 2): the monomial (Re ψ)(Im ψ)². -/
theorem L1_degree_three (a b : Nat)
    (heven : b % 2 = 0) (ha : 1 ≤ a) (hb : 1 ≤ b) (hdeg : a + b = 3) :
    a = 1 ∧ b = 2 := by
  omega

/-- Uniqueness packaged: the set of admissible (a, b) of degree 3 is {(1, 2)}. -/
theorem L1_unique (a b : Nat) :
    (b % 2 = 0 ∧ 1 ≤ a ∧ 1 ≤ b ∧ a + b = 3) ↔ (a = 1 ∧ b = 2) := by
  constructor
  · intro h; omega
  · intro h; omega

/-! ### The parity table -/

/-- ω = 2 Re ψ ~ (a, b) = (1, 0): ι-even. -/
theorem L1_table_omega : iotaSign 0 = 1 := by decide
/-- ψ − ψ̄ = 2i Im ψ ~ (0, 1): ι-odd. -/
theorem L1_table_odd : iotaSign 1 = -1 := by decide
/-- ψ( − ψ)² ~ (0, 2): ι-even. -/
theorem L1_table_odd_sq : iotaSign 2 = 1 := by decide
/-- Φ_QC = ωψ( − ψ)² ~ (1, 2): ι-even. -/
theorem L1_table_PhiQC : iotaSign 2 = 1 := by decide
/-- ι-sign is multiplicative in the Im-exponent: sign(₁b + ₂b) = sign ₁b · sign ₂b. -/
theorem L1_sign_mul (₁b ₂b : Nat) : iotaSign (₁b + ₂b) = iotaSign ₁b * iotaSign ₂b := by
  unfold iotaSign
  by_cases h1 : ₁b % 2 = 0 <;> by_cases h2 : ₂b % 2 = 0 <;> simp [h1, h2] <;> omega

end M2L1
```

M2_L2_master.lean — the dilaton term

```
/-!
# M2 — Certificate L2: the dilaton term of the Master Equation (Section 3)

Core toolchain, Lean 4.15.0, no Mathlib, no author axioms, zero `sorry`.

Certified, on an explicit finite model (a diagonal orthonormal frame withΦ
 depending on the flow parameter only; components as integers scaled byζ
```

```
2/( Φ²) to stay in ℤ):
* `L2_flow_components` — TΦ^_ss = Φ⃗½² and TΦ^_ii = −Φ⃗½², i.e. in doubled
                         units (1, −…−1,,1) with trace 1 − 11 = −10;
* `L2_trace`           — the trace g^{MN}TΦ^_{MN} = (1 − 12/2)∇Φ||² = −∇Φ5||²
                         in twelve dimensions (doubled: −10);
* `L2_kinetic_parity`  — ∇Φ||² is τ̂even: Φ ~ (Re ψ)(Im ψ)² has Im-exponent 2,
                         and the gradient does not change the Im-parity, so
                         the product of two gradients has even Im-exponent;
* `L2_kinetic_degree`  — ∇Φ||² has degree 6 in the components of ψ;
* `L2_competitors`     — the competing two-derivative terms fail the
                         vanishing conditions of Prop. parity: ∇ω||² ~ (Re ψ)^0
                         in the Im-exponent (does not vanish at ψ = ψ⃗),
                         ∇ψ||² has Re- and Im-exponents 0 (vanishes nowhere).
The analytic content (variation, Bianchi contraction) is the one-line
computation in the text; this certificate fixes the arithmetic the text relies on.
-/

namespace M2L2

/-- Doubled components of TΦ^ in the frame ∂(_s, e_1, …, e_11): index 0 is the flow. -/
def comp (i : Nat) : Int := if i = 0 then 1 else -1

theorem L2_flow_components : comp 0 = 1 ∧ ∀( i, 1 ≤ i → i ≤ 11 → comp i = -1) := by
  constructor
  · decide
  · intro i h1 _
    unfold comp
    have : i ≠ 0 := by omega
    simp [this]

/-- Trace in d = 12 (doubled): 1 + 11·−(1) = −10, i.e. (1 − d/2)·2 with d = 12. -/
theorem L2_trace : ((List.range 12).map comp).foldl (· + ·) 0 = -10 := by decide
theorem L2_trace_formula : (2 : Int) - 12 = -10 := by decide

/-- Monomial bookkeeping in (Re ψ, Im ψ): Φ ↦ (1,2). A gradient preserves the
    Im-exponent parity; the product of two gradients of (a,b) has Im-exponent
    2b − (0 or 2), even iff b even. -/
def PhiExp : Nat × Nat := (1, 2)
def gradSq (e : Nat × Nat) : Nat × Nat := (2*e.1, 2*e.2)   -- leading term ∂Φ()² ~ Φ²/length²

theorem L2_kinetic_parity : (gradSq PhiExp).2 % 2 = 0 := by decide
theorem L2_kinetic_degree : (gradSq PhiExp).1 + (gradSq PhiExp).2 = 6 := by decide

/-- Competitors: ω ~ (1,0), ψ ~ (1,0)+(0,1). ∇ω||² has Im-exponent 0 (b = 0 fails
    "vanishes at ψ = ψ⃗", which needs b ≥ 1); ∇ψ||² contains the monomial (0,0),
    which fails both vanishing conditions. -/
def omegaExp : Nat × Nat := (1, 0)
theorem L2_competitors :
    (gradSq omegaExp).2 = 0 ∧ ¬ (1 ≤ (gradSq omegaExp).2) ∧
    ((0:Nat), (0:Nat)) = (0,0) ∧ ¬ (1 ≤ (0:Nat)) := by decide

end M2L2
```

M2_L3_zeta.lean — the integers of ζ

```
/-!
# M2 — Certificate L3: the arithmetic face of ζ = π12 (Section 4 / WP4)

Core toolchain, Lean 4.15.0, no Mathlib, no author axioms, zero `sorry`.

What this certifies, and what it does not. ζ = π12 is derived in M2 §4 as the
total warp exponent ωk_ ωL_ = N π of the two-dimensional —EinsteinMaxwell
```

```
system on the pillowcase with N = χ(₈K) flux units (BPS completion, Dirac
quantisation, kink balance, –GaussBonnet normalisation κ₂² = 2/ωL_). That
derivation is analytic and is in the text. This file certifies only its
integer skeleton, with π carried as a formal unit (all angles in units of π):

* `L3_chi_K8`        – χ(₈K) = χℂ(P²)χ·(S²)χ·(T²ℤ₂/) = 3·2·2 = 12 (M1 Thm 7.6);
* `L3_deficit_each`  – a cone of angle π has deficit π2 − π = π;
* `L3_deficit_total` – four such cones total π4 = π2χ·(S²), χ = 2 (–GaussBonnet);
* `L3_zeta_units`    – ζπ/ = N = χ(₈K) = 12, and 12 = 4·3 = ππ(4/)χ·ℂ(P²):
                       the cone-deficit reading and the χ(₈K) reading are one
                       identity read in two orders (Künneth), not two routes.
-/

namespace M2L3

def chi_CP2    : Nat := 3   -- M1 Prop 7.2 (L4_chi_counts)
def chi_S2     : Nat := 2
def chi_pillow : Nat := 2   -- M1 Thm 5.7 (L2_chi_pillowcase)
def chi_K8     : Nat := chi_CP2 * chi_S2 * chi_pillow   -- M1 Thm 7.6 (L3_chi_K8)

theorem L3_chi_K8 : chi_K8 = 12 := by decide

/-- Angles in units of π: a full turn is 2, a cone point of the pillowcase is 1. -/
def fullTurn  : Nat := 2
def coneAngle : Nat := 1
def deficit   : Nat := fullTurn - coneAngle
def nCones    : Nat := 4

theorem L3_deficit_each : deficit = 1 := by decide
/-- Total deficit π4 equals π2χ·(pillowcase) with χ = 2 (–GaussBonnet, M1 Thm 5.7). -/
theorem L3_deficit_total : nCones * deficit = fullTurn * chi_pillow := by decide

/-- ζ in units of π equals the flux number N = χ(₈K) (tadpole, M2 §4). -/
def N_flux     : Nat := chi_K8
def zeta_units : Nat := N_flux

theorem L3_zeta_units : zeta_units = 12 := by decide
/-- The cone-deficit reading: (total deficit / π) × χℂ(P²) = 4·3 = 12. -/
theorem L3_cone_reading : nCones * deficit * chi_CP2 = zeta_units := by decide
/-- The two readings coincide identically by the Künneth factorisation. -/
theorem L3_one_identity :
    nCones * deficit * chi_CP2 = chi_CP2 * chi_S2 * chi_pillow := by decide

end M2L3
```

M2_L4_handedness.lean — handedness

```
/-!
# M2 – Certificate L4: handedness relative to the flow (Section 2 / WP2, R2.3)

Core toolchain, Lean 4.15.0, no Mathlib, no author axioms, zero `sorry`.
External input named, never a Lean axiom: an orientation-reversing element of
Pin anticommutes with the chirality operator in every signature
(–LawsonMichelsohn, Spin Geometry, I.§5).

Model. The Sol block is the suspension of the drive: (x, s) ~ (₀Mx, s+1) with₀
M = ((2,1),(1,1)) (M1 Thm 4.2). A symmetry reversing the flow has the form
(x, s) ↦ (Ax, −s) with A ∈ GLℤ(2,) and A ₀M ⁻A¹ = ₀⁻M¹. Its orientation on the
three-dimensional block is det A · −(1).

Certified:
* `L4_S_conj`        – the quarter-turn S conjugates ₀M to ₀⁻M¹ (M1 `L1_conj_M0_inv`,
```

```
                        re-certified here under the island rule);
* `L4_S_sq_fold`     — S² = −I: the square of the quarter-turn is M1's fold;
* `L4_S_det`         — det S = +1, so τ = (S, s ↦ −s) has total determinant −1:
                        it reverses the orientation of the block;
* `L4_swap_lines`    — in ℤ[φ], S carries the unstable eigenvector φ(, 1) of ₀M
                        to minus the stable eigenvector −(1, φ): the quarter-turn
                        exchanges growth and decay;
* `L4_eigen`         — ₀Mφ(,1) = φ²φ(,1) and ₀M₋φ(1,) = φ⁻²₋φ(1,), φ⁻² = 2 − φ;
* `L4_involution_det₋` any A with A² = I and A ₀M ⁻A¹ = ₀⁻M¹ has det A = −1, so
                        an *involutive* flow reversal preserves the block's
                        orientation; orientation reversal forces order four,
                        with square the fold.
Hence ψ⁻≡ τψ̂ is related to ψ by an odd element of Pin and has the opposite
chirality — in Riemannian signature, with no Lorentzian structure assumed.
-/

namespace M2L4

/-- 2×2 integer matrices ((a,b),(c,d)). -/
structure M2 where
  a : Int
  b : Int
  c : Int
  d : Int
deriving DecidableEq, Repr

namespace M2
def mul (p q : M2) : M2 :=
  ⟨p.a*q.a + p.b*q.c, p.a*q.b + p.b*q.d, p.c*q.a + p.d*q.c, p.c*q.b + p.d*q.⟩d
def det (p : M2) : Int := p.a*p.d - p.b*p.c
def I    : M2 := ⟨⟩1,0,0,1
def negI : M2 := ⟨⟩-1,0,0,-1
def M0   : M2 := ⟨⟩2,1,1,1
def M0inv: M2 := ⟨⟩1,-1,-1,2
def S    : M2 := ⟨⟩0,1,-1,0   -- quarter-turn
def Sinv : M2 := ⟨⟩0,-1,1,0
end M2
open M2

theorem L4_M0_inv : mul M0 M0inv = I ∧ mul M0inv M0 = I := by decide
theorem L4_S_inv  : mul S Sinv = I := by decide
/-- S ₀M ⁻S¹ = ₀⁻M¹ (re-certification of M1 `L1_conj_M0_inv`). -/
theorem L4_S_conj : mul (mul S M0) Sinv = M0inv := by decide
/-- S² = −I: the quarter-turn squares to the fold. -/
theorem L4_S_sq_fold : mul S S = negI := by decide
theorem L4_S_det : det S = 1 := by decide
/-- Total orientation of τ = (S, s ↦ −s) on the Sol block: det S · −(1) = −1. -/
theorem L4_tau_reverses : det S * (-1) = -1 := by decide

/-! ### The golden ring ℤ[φ], φ² = φ + 1, and the eigen-directions -/

/-- Elements a + φb of ℤ[φ]. -/
structure Zphi where
  a : Int
  b : Int
deriving DecidableEq, Repr

namespace Zphi
def add (x y : Zphi) : Zphi := ⟨x.a+y.a, x.b+y.⟩b
/-- (a+φb)(c+φd) = ac + (ad+bc)φ + bd φ² = (ac+bd) + (ad+bc+bd)φ. -/
def mul (x y : Zphi) : Zphi := (x.a*y.a + x.b*y.b, x.a*y.b + x.b*y.a + x.b*y.)b
def ofInt (n : Int) : Zphi := ⟨n⟩,0
def phi   : Zphi := ⟨⟩0,1
```

```
def phi2  : Zphi := ⟨⟩1,1      -- φ² = 1 + φ
def phiInv2 : Zphi := ⟨⟩2,-1   -- φ⁻² = 2 − φ
def neg (x : Zphi) : Zphi := ⟨-x.a, -x.⟩b
end Zphi
open Zphi

/-- Vectors in ℤφ[]². -/
structure V where
  x : Zphi
  y : Zphi
deriving DecidableEq, Repr

def act (m : M2) (v : V) : V :=
  ⟨Zphi.add (Zphi.mul (ofInt m.a) v.x) (Zphi.mul (ofInt m.b) v.y),
   Zphi.add (Zphi.mul (ofInt m.c) v.x) (Zphi.mul (ofInt m.d) v.y)⟩
def smul (l : Zphi) (v : V) : V := ⟨Zphi.mul l v.x, Zphi.mul l v.⟩y
def negV (v : V) : V := ⟨Zphi.neg v.x, Zphi.neg v.⟩y

def vU : V := ⟨phi, ofInt ⟩1        -- unstable direction φ(, 1)
def vS : V := ⟨ofInt (-1), ⟩phi     -- stable direction −(1, φ)

theorem L4_phi2 : Zphi.mul phi phi = phi2 := by decide
theorem L4_phi_inv2 : Zphi.mul phi2 phiInv2 = ofInt 1 := by decide
/-- ₀M v_u = φ² v_u and ₀M v_s = φ⁻² v_s. -/
theorem L4_eigen : act M0 vU = smul phi2 vU ∧ act M0 vS = smul phiInv2 vS := by decide
/-- The quarter-turn exchanges the two lines: S v_u = −v_s, S v_s = v_u. -/
theorem L4_swap_lines : act S vU = negV vS ∧ act S vS = vU := by decide

/-! ### Involutive flow reversals preserve orientation -/

/-- If A² = I and A ₀M = ₀⁻M¹ A (i.e. A ₀M ⁻A¹ = ₀⁻M¹), then det A = −1. -/
theorem L4_involution_det (A : M2) (hsq : M2.mul A A = I) (hconj : M2.mul A M0 = M2.mul M0inv A) :
    det A = -1 := by
  obtain ⟨a, b, c, ⟩d := A
  simp only [M2.mul, I, M0, M0inv, M2.mk.injEq, det] at hsq hconj ⊢
  obtain ⟨h1, h2, h3, ⟩h4 := hsq
  obtain ⟨k1, k2, k3, ⟩k4 := hconj
  -- h2 : a*b + b*d = 0, h3 : c*a + d*c = 0
  have hb : b * (a + d) = 0 := by rw [Int.mul_add, Int.mul_comm b a]; omega
  have hc : c * (a + d) = 0 := by rw [Int.mul_add, Int.mul_comm c d]; omega
  rcases Int.mul_eq_zero.mp hb with hb0 | had
  · subst hb0
    rcases Int.mul_eq_zero.mp hc with hc0 | had
    · subst hc0
      -- A = diag(a,d); conjugation gives 2a = a, so a = 0, contradicting a² = 1
      have ha : a = 0 := by omega
      subst ha
      simp at h1
    · have hd : d = -a := by omega
      subst hd
      have h1' : a * a = 1 := by simpa using h1
      have : a * -a = -(a * a) := Int.mul_neg a a
      omega
  · have hd : d = -a := by omega
    subst hd
    have : a * -a = -(a * a) := Int.mul_neg a a
    omega

end M2L4
```

M2_L5_warp.lean — the warp exponent

```
/-!
# M2 — Certificate L5: the warp exponent and ζ = π12 (Section 4 / WP4)

Core toolchain, Lean 4.15.0, no Mathlib, no author axioms, zero `sorry`.
Ports the integer content of the fossil certificates P2_L1 (Euler
characteristics) and P2_L2 (`zeta_coefficient`, `single_flux_case`), and adds
the balance identity and the corner action of τ.

Certified π( carried as an abstract integer symbol x throughout):
* `L5_chi_pillowcase` — orbifold count 0 + 4·(1/2) = 2 (doubled: 0 + 4 = 4 = 2·2);
* `L5_deficit`        — four corners × deficit π = π4 = π2χ·, χ = 2;
* `L5_chi_K8`         — 3·2·2 = 12;
* `L5_FD_check`       — N/2 over the fundamental domain, doubled over the cover: 2·(N/2) = N for even N,
     and 2N = 2N in general (stated with the halving cleared);
* `L5_balance`        — from 2k = κπ²N and κ² = 2/L (cleared: 2kL = 2πN) follows kL = πN;
* `L5_zeta`           — ζ = χ(₈K)π· = π12 at coefficient level;
* `L5_single_flux`    — N = 1 gives exponent π;
* `L5_tau_corners`    — the quarter-turn S on the half-lattice (coordinates doubled, mod 2) fixes ₁P =
     (0,0) and ₄P = (1,1) and exchanges ₂P = (1,0) ↔ ₃P = (0,1).
-/

namespace M2L5

theorem L5_chi_pillowcase : (0:Nat) + 4 * 1 = 2 * 2 := by decide
theorem L5_deficit (x : Int) : 4 * x = 2 * x * 2 := by omega
theorem L5_chi_K8 : (3:Nat) * 2 * 2 = 12 := by decide

/-- Fundamental-domain check: the FD carries N/2 units, the cover 2·(N/2) = N. -/
theorem L5_FD_check (N : Int) (h : N % 2 = 0) : 2 * (N / 2) = N := by omega
/-- Cleared form, any N: twice the FD integral equals the cover integral. -/
theorem L5_FD_check_cleared (N : Int) : 2 * N = N + N := by omega

/-- Balance: 2k = κπ²N with κ² = 2/L. Cleared of the division (multiply by L):
    2kL = 2πN, hence kL = πN. -/
theorem L5_balance (k L N x : Int) (h : 2 * (k * L) = 2 * (N * x)) : k * L = N * x := by
  omega

theorem L5_zeta (x : Int) : (3 * 2 * 2) * x = 12 * x := by omega
theorem L5_single_flux (x : Int) : 1 * x = x := by omega

/-- Quarter-turn S = ((0,1)−,(1,0)) on doubled corner coordinates, reduced mod 2. -/
def S_mod2 (p : Int × Int) : Int × Int := ((p.2) % 2, ((-p.1) % 2 + 2) % 2)
def P1 : Int × Int := (0,0)
def P2 : Int × Int := (1,0)
def P3 : Int × Int := (0,1)
def P4 : Int × Int := (1,1)

theorem L5_tau_corners :
    S_mod2 P1 = P1 ∧ S_mod2 P4 = P4 ∧ S_mod2 P2 = P3 ∧ S_mod2 P3 = P2 := by decide

end M2L5
```

M2_L6_omega.lean — the shape of ω

```
/-!
# M2 — Certificate L6: the shape of ω and the sign bookkeeping (Section 5 / WP5)

Core toolchain, Lean 4.15.0, no Mathlib, no author axioms, zero `sorry`.

Certified (trigonometric content reduced to algebra in s = ϑsin², c = ϑcos²,
s + c = 1, cleared to thirds):
```

```
* `L6_Phi_coefficient`  — Φ = ωψ−ψ⃒|² = (2r ϑcos)|2ir ϑsin|² = 8 r³ ϑcos ϑsin²:
                          the coefficient 2·(2²) = 8 (PI ruling 9 Nov 2026: modulus squared);
* `L6_Phi_extremum`     — d/ϑd(ϑcos ϑsin²) = ϑsin(2ϑcos² − ϑsin²) vanishes at
                          2c = s, i.e. s = 2/3, c = 1/3 (ϑtan² = 2);
* `L6_sign`             — if ψ−ψ⃒|² = q > 0 and Φ = ω·q ≥ 0 then ω ≥ 0: the
                          dilaton's positivity on the decaying-warp solutions
                          fixes the sign of Re ω wherever the sheets are
                          distinct (the half-circle); flipping ψ flips both
                          (`L6_flip`);
* `L6_hump_zeros`       — the circular-normal mean ⟨ω⟩ = 2 r̄ A(c) ϑ₀cos vanishes
                          at A = 0 (uniform) and at ϑ₀cos = 0 (locked), as the
                          product identity it is.
-/

namespace M2L6

/-- 8 = 2 · (2²): ω = 2r cos, ψ−ψ⃒|² = |2 i r sin|² = 4 r² sin². -/
theorem L6_Phi_coefficient : (2:Int) * (2 * 2) = 8 := by decide

/-- Extremum of cos·sin²: 2cos² = sin². With s + c = 3 (thirds), 2c = s ⇒ s = 2, c = 1. -/
theorem L6_Phi_extremum (s c : Int) (h1 : s + c = 3) (h2 : 2 * c = s) : s = 2 ∧ c = 1 := by
  omega

/-- Sign bookkeeping: q > 0 and ω·q ≥ 0 ⇒ ω ≥ 0. -/
theorem L6_sign (w q : Int) (hq : 0 < q) (h : 0 ≤ w * q) : 0 ≤ w := by
  by_cases hw : 0 ≤ w
  · exact hw
  · exfalso
    have hw' : w < 0 := by omega
    have : w * q < 0 := Int.mul_neg_of_neg_of_pos hw' hq
    omega

/-- Flipping the sign of ψ flips ω and Φ together: −(w)·q = −(w·q). -/
theorem L6_flip (w q : Int) : (-w) * q = -(w * q) := Int.neg_mul w q

/-- ⟨ω⟩ = 2 r̄ · A · ϑ₀cos vanishes when A = 0 or ϑ₀cos = 0. -/
theorem L6_hump_zeros (r A k : Int) : (A = 0 ∨ k = 0) → 2 * r * A * k = 0 := by
  intro h
  rcases h with h | h <;> simp [h]

end M2L6
```

M2_L7_module.lean — the Clifford module

```
/-!
# M2 — Certificate L7: the Clifford module and the module trace (Sections —23)

Core toolchain, Lean 4.15.0, no Mathlib, no author axioms, zero `sorry`.
Adopted from the τ₂ programme's ruling R-F1Φ() (24 Oct 2026) and re-certified
in M2's tree; the external input is Bott periodicity for the real Clifford
algebras, Cl(0,n+8) ≅ Cl(0,n) ⊗ ℝ(16), with the table Cl(0,4) ≅ ℍ(2).

Certified:
* `L7_bott`           — 12 = 4 + 8, 12 % 8 = 4: Cl(0,12) ≅ Cl(0,4) ⊗ ℝ(16) ≅ ℍ(2·16) = ℍ(32);
* `L7_module_dims`    — the irreducible module ℍ³² has real dimension 128,
                        complex dimension 64, quaternionic dimension 32;
                        the algebra ℍ(32) has real dimension 4·32² = 4096 = 2¹²;
* `L7_three_readings` — for a module-symmetric configuration with per-component
                        value c, the three readings of ζ∇Φ∇Φ are c (one
                        component), 32c (trace of the square), 1024c (square
                        of the trace); only the trace is the module inner product;
* `L7_trace_uniform`  — the trace of a uniform vacuum is 32 × one component;
```

```
* `L7_imaginary_square_real` — for a pure imaginary quaternion q = bi + cj + dk,
                         q² = −(b² + c² + d²) is real: hence ψ−ψ()² contributes
                         no imaginary direction, and ω is its unique carrier.
-/

namespace M2L7

theorem L7_bott : (12:Nat) = 4 + 8 ∧ 12 % 8 = 4 ∧ (2:Nat) * 16 = 32 := by decide

theorem L7_module_dims :
    (4:Nat) * 32 = 128 ∧ (2:Nat) * 32 = 64 ∧ (4:Nat) * (32 * 32) = 4096 ∧ 2 ^ 12 = 4096 := by
  decide

/-- Three readings of the kinetic term for a uniform configuration of 32 components. -/
def oneComponent (c : Nat) : Nat := c
def traceOfSquare (c : Nat) : Nat := 32 * c
def squareOfTrace (c : Nat) : Nat := (32 * 32) * c

theorem L7_three_readings (c : Nat) :
    oneComponent c = c ∧ traceOfSquare c = 32 * c ∧ squareOfTrace c = 1024 * c := by
  unfold oneComponent traceOfSquare squareOfTrace
  refine ⟨rfl, rfl, ?_⟩
  omega

theorem L7_trace_uniform (c : Nat) : (List.replicate 32 c).foldl (· + ·) 0 = 32 * c := by
  have h : ∀ (n acc : Nat), (List.replicate n c).foldl (· + ·) acc = acc + n * c := by
    intro n
    induction n with
    | zero => intro acc; simp
    | succ k ih =>
      intro acc
      simp [List.replicate_succ, List.foldl_cons, ih, Nat.succ_mul]
      omega
  simpa using h 32 0

/-- Quaternion multiplication on (a,b,c,d) = a + bi + cj + dk. -/
structure Q where
  a : Int
  b : Int
  c : Int
  d : Int
deriving DecidableEq, Repr

def Q.mul (p q : Q) : Q :=
  ⟨p.a*q.a - p.b*q.b - p.c*q.c - p.d*q.d,
   p.a*q.b + p.b*q.a + p.c*q.d - p.d*q.c,
   p.a*q.c - p.b*q.d + p.c*q.a + p.d*q.b,
   p.a*q.d + p.b*q.c - p.c*q.b + p.d*q.⟩a

/-- The square of a pure imaginary quaternion is the real number −(b²+c²+d²). -/
theorem L7_imaginary_square_real (b c d : Int) :
    Q.mul ⟨0,b,c,⟩d ⟨0,b,c,⟩d = ⟨-(b*b + c*c + d*d), 0, 0, ⟩0 := by
  simp only [Q.mul, Q.mk.injEq]
  refine ⟨?_, ?_, ?_, ?⟩_
  · omega
  · rw [Int.mul_comm c d]; omega
  · rw [Int.mul_comm b d]; omega
  · rw [Int.mul_comm b c]; omega

end M2L7
```

B The labelling of the corners, and the symbols of the series

From M1 (reproduced). The four corners are named by position, $P_1 = (0,0)$, $P_2 = (\frac{1}{2},0)$, $P_3 = (0,\frac{1}{2})$, $P_4 = (\frac{1}{2},\frac{1}{2})$ in coordinates $(\theta_\psi, \theta_\omega)$, and by algebra, $C_0 = \mathbb{R}$, C_i, C_j, C_k. Throughout the programme the dictionary is: $P_1 = C_0$ (fixed by the drive; ι acts as conjugation); $P_3 = C_i$ (the ω half-period); $P_2 = C_j$ (the ψ half-period); $P_4 = C_k$ (the corner at which the Spin^c lift carries the sign -1). The assignment of j and k between P_2 and P_3 is a convention and carries no content (M1 Appendix B).

Introduced in M2.

Symbol	Meaning	Where defined
$\psi,\ \bar{\psi} = \hat{\tau}\psi$	the sheet field, a section of the Clifford module $M \cong \mathbb{H}^{32}$, and its shadow	Def. 2.2
M, $\langle\cdot,\cdot\rangle_M$	irreducible $\mathrm{Cl}(0,12)$-module; its inner product (trace)	Def. 2.2, §3.1
$\mathcal{A}$, $\mathcal{C}$	bundle of fibre algebras; corner-algebra subbundle (the corner reading)	Def. 2.2
$\tau,\ \hat{\tau}$	flow-reversing involution of A2 (τ^2 = fold on the cover); its lift	§2.5
S, v_u, v_s, $\kappa = \log\varphi^2$	quarter-turn; unstable/stable directions of M_0; Sol rate	§2.5
$\omega = \psi + \bar{\psi}$	the Coherent Field; $\hat{\tau}$-even, quaternion-valued, carries the imaginary direction	Def. 2.3
$\psi - \bar{\psi}$	the odd sheet variable; $\hat{\tau}$-odd	Def. 2.3
$\Phi_{\mathrm{QC}} = \omega\lvert\psi - \bar{\psi}\rvert^2$	the Quantum Coherence Dilaton, componentwise in M	Prop. 2.4
A, $w = e^A$	the A2 warp exponent and warp factor on the base	Def. 2.5
k_ω	decay rate of A along θ_ω	Section 4
F	curvature of the Spin^c determinant line	Def. 2.5
κ_2	two-dimensional gravitational coupling on the base	Section 4
$T^{\Phi_{\mathrm{QC}}}_{MN}$	stress tensor of Φ_{QC}, the module trace of the scalar form	Section 3
ζ	coupling of Φ_{QC}: coefficient of $T^{\Phi_{\mathrm{QC}}}_{MN}$; equals the warp exponent per half-period	Sections 3–4
$G_{\mathrm{eff}},\ \Lambda_{\mathrm{eff}}$	effective gravitational and vacuum couplings	Section 3

Later papers of the series reproduce this table and extend it.

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